Low-Risk and Coskewness-Based Stock Selection: The Incremental Value of Portfolio Optimization

Team #4

***Abstract*—We evaluate portfolio allocation following stock selection based on low beta, low volatility, and negative coskewness. Nineteen rules use the same 30 stocks, rolling 36-month estimates, and position constraints during February 2003–December 2020. With observed CRSP returns and 20-basis-point round-trip costs, market-capitalization weighting achieves the highest compound annual growth rate (CAGR), 14.05%, and Sharpe 1.110. All allocations exceed the S&P 500 price benchmark's CAGR of 8.61%, as well as the CRSP and Fama–French total-return benchmarks' 10.86% and 11.54%. The S&P comparison excludes benchmark dividends and is interpreted separately. The Sharpe gap against the FF market is nominally significant over the full sample, but not during 2012–2020. Ledoit–Wolf identity shrinkage has the highest Sharpe, 1.114, but its advantage is statistically indistinguishable from zero. No allocation mean-return difference from market-capitalization weighting survives multiple-comparison adjustment. Exposure and cost decompositions show how reweighting changes the portfolio. Evidence of incremental allocation benefits remains limited and conditional on historical selection.**

***Index Terms*—Portfolio selection, low-risk investing, coskewness, covariance estimation, transaction costs, backtesting.**

I. Overview

A. Motivation

PORTFOLIO performance depends on both security selection and weighting. Minimum-variance optimization following low-risk stock selection operates within an already restricted investment set. Its incremental benefit must therefore be assessed by applying alternative weighting rules to the same securities under common constraints and return conventions.

We study a strategy combining low beta, low volatility, and negative coskewness. The first two characteristics restrict market and total risk; the third reflects potential compensation for systematic skewness exposure. All allocation rules use the same 30 stocks each month, including constituents assigned zero weight by an optimizer. Differences between portfolios thus reflect weighting conditional on historical selection.

The primary strategy benchmark is the original market-capitalization-weighted portfolio with a $2/30$ position limit (6.67%, rounded), labeled "Market-cap weight in tables and figures; equal weighting provides a second reference. Common position limits and constituents isolate allocation effects. S&P 500 is the primary market benchmark, with CRSP and FF market returns retained as complementary comparisons. We distinguish price from total returns and descriptive growth differences from paired statistical inference.

We also examine how weighting changes characteristic exposures, concentration, and turnover. All strategies use observed CRSP total returns, including termination months, and a round-trip cost of 20 basis points.

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B. Related Literature and Research Questions

Mean–variance analysis determines portfolio weights from expected returns and return covariances [1]. Estimation error can erode the gains from optimization: DeMiguel, Garlappi, and Uppal [2] document the difficulty of consistently outperforming equal weighting out of sample. Linear shrinkage combines sample covariance with a structured target [3], whereas nonlinear shrinkage adjusts sample eigenvalues [4]. These approaches motivate our covariance specifications.

Leverage constraints may increase demand for high-beta securities and help explain the low-beta anomaly [5]. Our long-only strategy uses a smoothed beta characteristic rather than replicating Betting Against Beta. Preferences over systematic skewness likewise motivate coskewness selection [6], but characteristic-based selection alone does not establish a priced risk premium.

Searching across signals, portfolio sizes, and implementation choices introduces data-snooping risk [7]. Common constituents prevent allocation-specific selection changes but do not remove bias from the earlier strategy search. Subperiod results describe temporal variation; they are not independent validation samples.

We ask whether alternative weighting improves net performance, whether differences arise from mean returns or volatility, and how trading costs affect the comparison. Paired mean-return tests and Sharpe-difference confidence intervals evaluate the incremental outcomes.

C. Overall Investment Framework

Figure 1 summarizes the monthly procedure. Weights chosen at the end of month t use information available through t and earn returns in month $t+1$. The estimation window precedes this holding period. Return definitions, costs, benchmarks, and evaluation periods are common to all rules.

II. Investment Universe

A. Asset Pool and Eligibility

The documented universe comprises common stocks listed on the NYSE, AMEX, and NASDAQ, regardless of issuer domicile. Eligibility excludes SPAC-like firms, stocks below the contemporaneous NYSE 20th-percentile capitalization breakpoint, and securities terminated before rebalancing. These filters limit exposure to very small firms and apparently low-risk securities dominated by cash or infrequent trading. The NYSE breakpoint differs from excluding the smallest quintile across all exchanges.

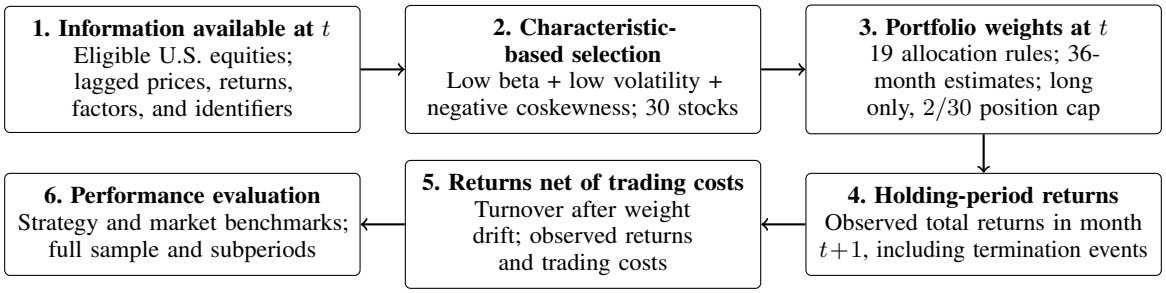


Fig. 1. Monthly investment procedure. Steps 1–2 document selection; Steps 3–6 compare allocations on common constituents. Regimes enter evaluation only.

B. Data Sources and Sample

The sample comprises 6,450 stock-month records: 30 securities per month over 215 holding months, February 2003–December 2020. Each record includes the preceding rebalancing month, three characteristic scores, their composite, selection rank, and original weight. Holding months follow rebalancing by one month without gaps. PERMNO is the matching key; tickers are descriptive.

We combine this file with the course price panel, original CRSP monthly returns, termination dates, and Fama–French factors. Observed CRSP returns cover all 6,450 holding observations and 28,470 distinct security-months in the rolling 36-month estimation windows. No constituents are removed or substituted. Capitalization and SIC fields are missing for 1,088 observations in the filtered metadata extract, so the benchmark retains its recorded weights.

C. Return Pre-processing and Terminal Events

Monthly total returns come from the original CRSP CIZ `MthRet` field, matched by PERMNO and month [8]. The loader checks key uniqueness and agreement with overlapping cleaned records, then restores omitted observations. No additional delisting return, price-return substitution, winsorization, or loss imputation is applied. Missing required estimation or holding returns stop the experiment.

The cleaned extract omits 1,109 selected stock-months: 1,086 ordinary observations for non-U.S.-incorporated issuers and 23 termination observations no longer classified as active common shares. All returns exist in the original file. The pattern suggests overly restrictive metadata filtering, although the unavailable cleaning code prevents verification of the exact filter. Returns are therefore retrieved by security-month key without contemporaneous domicile or active-status restrictions.

The 23 restored terminal returns range from -2.4164% to $+9.1320\%$ and correspond to 20 mergers and three exchange or reclassification events. They replace erroneous losses imposed in earlier results. After the observed terminal return, proceeds remain in cash until rebalancing. Original files and superseded outputs are archived separately.

D. Estimation, Validation, and Evaluation Timing

Allocation inputs use rolling 36-month windows ending at t; performance is measured at $t+1$. The first window, February 2000–January 2003, precedes the February 2003 holding month. Covariance rules estimate second moments; Bayes–Stein also estimates means. Allocation parameters are fixed, with no separate validation sample used to select a rule. Returns are out of sample relative to rolling input estimates, but not necessarily relative to the historical choice of selection strategy.

The original research reports score regressions and portfolio-size selection under five-year rolling, ten-year fixed, and ten-year rolling schemes (Section III). These are not independently replicated: the selected-stock file lacks the full candidate universe needed to reconstruct their estimation and validation samples. An independent test of signal discovery would require prespecified rules and previously unused data. The present study uses only existing project datasets. The February 2003–December 2011 and January 2012–December 2020 comparisons below are retrospective temporal diagnostics, not verified training and untouched test samples. The available records do not establish that the selection specification was fixed before 2012.

III. Portfolio Framework

A. Economic Design of the Selection Rule

1) Economic Motivation for the Selection Characteristics: The selection rule combines market- and total-risk restrictions with potential systematic-skewness compensation. Its objective is risk-adjusted performance: a characteristic can improve the Sharpe ratio by reducing volatility despite weak raw-return predictability. Higher expected returns likewise need not improve the ratio if risk rises sufficiently.

The original search considered liquidity, issuance, momentum, and low-risk characteristics. Preliminary 52-week-high/low-trading-activity portfolios showed high turnover and sensitivity to microcapitalization stocks, motivating greater emphasis on feasibility and risk reduction. This history explains the specification but also exposes it to selection bias; it does not establish optimality among alternatives.

Low beta and low volatility constrain distinct risks. Under $r_i = \alpha_i + \beta_i m + \varepsilon_i$, with market-orthogonal residuals, $\mathrm{Var}(r_i) = \beta_i^2 \mathrm{Var}(m) + \mathrm{Var}(\varepsilon_i)$. A low-beta stock may retain substantial idiosyncratic risk. Combining the characteristics excludes such stocks. Leverage aversion [5] and benchmark-related limits to arbitrage [9] motivate low-risk pricing, without predicting outperformance in every regime.

Negative coskewness may command compensation when investors prefer positive aggregate portfolio skewness [6]. It does not itself provide downside protection: the original

single-characteristic portfolio has high volatility and a severe drawdown. Combining coskewness with low-risk characteristics seeks potential risk compensation subject to risk reduction.

2) Measurement Assumptions and Their Consequences: Daily returns produce monthly beta estimates, bounded at -3 and 5 and averaged over the preceding 12 months with at least nine estimates required. Stocks are ranked by $0.6\overline{\beta}+0.4$. Bounding limits extreme estimates; averaging trades responsiveness for stability. The positive affine transformation preserves ranks:

$$\operatorname{rank}(0.6\overline{\beta} + 0.4) = \operatorname{rank}(\overline{\beta}). \tag{1}$$

Thus, rank stabilization can arise from estimation, bounding, or averaging, but not from the affine transformation.

The volatility score is the negative sample standard deviation of the preceding 12 monthly returns, all required. Its predictive relevance depends on volatility persistence, not a specified direction of expected returns. Monthly sampling also gives it different sensitivity to within-month variation than daily-return beta.

The coskewness characteristic is

$$C_{i,t} = -\frac{\mathbb{E}[(x_i - \overline{x}_i)(m - \overline{m})^2]}{\sigma_{x_i}\sigma_m^2}, \tag{2}$$

where stock and market excess-return moments use the preceding 60 months, with at least 36 required. This longer window improves third-moment precision at the cost of responsiveness. Squared market deviations include both signs, so coskewness is not a downside-only measure. For centered returns satisfying $x_i = \beta_i m + \varepsilon_i$,

$$\mathbb{E}[x_i m^2] = \beta_i \mathbb{E}[m^3] + \mathbb{E}[\varepsilon_i m^2]. \tag{3}$$

Market skewness therefore induces a beta-related component that volatility normalization does not remove. Residual coskewness would define a different characteristic; low unconditional rank correlation alone does not establish independent risk exposures.

3) Evidence on Individual and Combined Characteristics: Table I reproduces the research memorandum's characteristic comparisons under its original return and cost conventions [10]. Low beta and low volatility have Sharpe ratios of 0.70 and 0.81; combining them gives 0.83. Adding coskewness to either raises the ratio to 0.92 or 0.86. The three-characteristic portfolio has volatility 11.4%, Sharpe 1.11, and MDD -29.0%, with the latter two in bold. Coskewness alone has Sharpe 0.60 and the underlined MDD of -64.0%. These descriptive results motivate combination rather than a defensive interpretation of coskewness alone.

Reported rank correlations are 0.55 for beta–volatility scores, -0.03 for beta–coskewness, and 0.04 for volatility–coskewness. Coskewness thus contributes less-correlated rankings, not evidence of an independent premium. All comparisons share the original sample and specification search.

Decile results distinguish risk-adjusted performance more clearly than raw returns. The highest- and lowest-scoring deciles report annual returns of 13.9% and 12.3%, but a monthly return-difference statistic of only 0.27. Their Sharpe ratios are 1.02 and 0.37, and betas 0.72 and 1.45. We use score ordering because D1/D10 labels vary across records. Reported three-factor alpha differences remain model-dependent and do not distinguish compensation for omitted risk from mispricing.

4) Equal Weighting of Characteristic Scores: The three percentile ranks are averaged, with higher scores indicating lower beta, lower volatility, and more negative coskewness. Ranks standardize units and limit outlier influence, but discard magnitude information and impose monotonic ordering. Equal score weights reduce estimation demands and preserve each characteristic's intended direction; they do not imply independence or equal risk contributions.

Expanding-window return regressions report coefficients of approximately -0.046, -0.049, and $+0.494$ monthly percentage points across the three score ranges, with statistics -0.13, -0.16, and 2.85. By December 2016, coskewness represents approximately 96% of absolute coefficient weight. The resulting Sharpe ratio is 0.25, versus 0.26 for a 60-month regression. Nonnegative coefficients raise it to 0.74, still below 1.11 for equal scores. Predicting raw returns does not explicitly reward the volatility reduction sought by low-risk selection.

Volatility-normalized regressions yield positive coefficients and Sharpe ratios around 0.96–0.97. This is closer to the risk-adjusted objective, but ignores covariance and costs and is not equivalent to maximizing portfolio Sharpe. The normalization timing cannot be verified from available records. Equal score weights therefore remain a parsimonious specification, not an established optimum.

5) Portfolio Size, Retention Rules, and Industry Constraints: The portfolio holds $K = 30$ stocks. Incumbents may remain while ranked within 45; other places go to the highest-ranked eligible stocks. This buffer reduces rank-induced turnover, although the selected-stock file cannot isolate its cost savings from the return effects of retaining lower-ranked stocks. At most nine stocks may belong to one two-digit SIC industry. This count limit differs from a 30% industry-weight cap. Individual portfolio weights are capped at $2/K$.

Reported full-sample Sharpe ratios vary little for $K \geq 25$ (approximately 1.03–1.11), while annual selection produces K from 10 to 100. The five-year rolling procedure yields Sharpe 0.87 for 2008–2020, versus 0.64 for the market. For 2013–2020, a fixed ten-year sample selects $K = 15$ and yields 0.86 versus market 1.04; the rolling ten-year scheme yields 1.02. These sensitivities preclude treating $K = 30$ as uniquely optimal.

6) Implications for the Allocation Stage: Within the restricted universe, minimum variance may increase low-volatility exposure while reducing coskewness exposure; mean-based optimization may favor high estimated returns; trading penalties may reduce costs. We assess these mechanisms through portfolio-weighted scores, concentration, fitted volatility, and gross-return/cost decompositions.

Evidence on characteristic combinations and portfolio size comes from the memorandum; allocation results and diagnostics are computed here. Missing daily beta inputs, full-universe scores, and selection code prevent upstream replication. Recorded composite scores match their equal-weighted components within 6.67×10^{-7}, but this arithmetic check does not validate the original ranks or unselected stocks.

TABLE I
CHARACTERISTIC PORTFOLIOS REPORTED IN THE ORIGINAL MEMORANDUM, NOT INDEPENDENTLY REESTIMATED. RETURN, VOLATILITY, AND MDD ARE PERCENTAGES; ANNUAL RETURN FOLLOWS THE ORIGINAL CONVENTION. B/V/C DENOTE LOW BETA, LOW VOLATILITY, AND NEGATIVE COSKEWNESS. BOLD MARKS COMBINED-PORTFOLIO SHARPE AND MDD; UNDERLINING MARKS COSKEWNESS-ONLY MDD.

Selection score	Report annual return	Volatility	Sharpe	MDD	Beta
B	12.2	14.8	0.70	-35.2	0.68
V	10.8	11.1	0.81	-36.8	0.64
C	15.6	22.3	0.60	<u>-64.0</u>	1.16
B + V	10.2	10.3	0.83	-33.0	0.55
B + C	15.0	13.9	0.92	-33.0	0.74
V + C	13.5	13.4	0.86	-40.7	0.80
B + V + C	14.8	11.4	**1.11**	**-29.0**	0.63

B. Mathematical Specification and Common Constraints

Let $X_t \in \mathbb{R}^{T\times N}$ contain returns for the constituents selected at t, with $T = 36$ and $N = 30$. All observations precede the holding period. Define $\bar{x} = T^{-1}X_t'\mathbf{1}$, $Y = X_t - \mathbf{1}\bar{x}'$, $S = Y'Y/(T-1)$, and $S_0 = Y'Y/T$. Their scaling affects shrinkage and mean estimation, although a common positive covariance multiplier leaves pure GMVP weights unchanged. Optimization inputs use monthly decimal units.

The feasible set is the capped simplex

$$\mathcal{W}_u = \{w \in \mathbb{R}^N : \mathbf{1}'w = 1,\ 0 \le w_i \le u\}, \qquad u = 2/N. \tag{4}$$

The feasible set is nonempty if and only if $Nu \ge 1$, compact, and convex. Positive semidefinite covariance ensures existence and convexity; positive definiteness ensures a unique minimizer. Common constraints isolate differences in covariance assumptions and objectives. The 19 rules include two benchmarks and a redundant norm-constrained GMVP, rather than 19 distinct statistical models.

1) The Common Minimum-Variance Problem: Equal weighting sets $w_i = 1/N$. The market-capitalization-weighted benchmark uses the weights recorded in the historical selection file, corrected only for six-decimal rounding by projection onto $\mathcal{W}_u$. The recorded weights remain the backtest inputs; an independent check against restored formation-date capitalizations is reported in Section IV-C8. For any covariance estimator $\widehat{\Sigma}$, the GMVP solves the constrained Markowitz problem [1]

$$\min_{w\in\mathcal{W}_u} f(w), \qquad f(w) = w'\widehat{\Sigma}w. \tag{5}$$

Without inequality constraints its solution is $\widehat{\Sigma}^{-1}\mathbf{1}/(\mathbf{1}'\widehat{\Sigma}^{-1}\mathbf{1})$. Applying a cap to that expression after optimization generally does not solve (5); the inequalities must enter the optimization problem. Sample GMVP uses S, estimating 465 distinct covariance elements from only 36 observations. Instability of small sample eigenvalues can consequently have substantial effects on weights.

C. Review of the Allocation Rules

Eight GMVP specifications differ in covariance estimation. Inverse-volatility and inverse-variance rules use marginal risk alone; risk budgeting, diversification, estimated Sharpe, and CVaR change the objective. Mixing, norm restrictions, and trading penalties regularize different aspects of allocation. Market-capitalization and equal weighting provide benchmarks. These categories are neither statistically independent nor necessarily distinct in their feasible sets.

Each entry links an allocation in Table II to its literature, assumptions, implementation, and observed behavior. Diagnostics use the common baseline: observed CRSP returns and 20 basis points round trip. They describe allocation outcomes without identifying causal estimator or characteristic effects. Sections III-D–III-F provide formulas and numerical verification.

For direct replication, each entry below includes a numbered allocation procedure. The procedures operate at rebalancing date t on the same ordered constituent set I_t, complete 36×30 return matrix X_t, and cap $u = 2/30$. The common inputs Y, S, and S_0 are defined in Section III-B; returns are monthly decimals. $[z]_a^b = \min\{b, \max\{a, z\}\}$ denotes scalar truncation, and $\Pi_{\mathcal{W}_u}$ denotes Euclidean projection onto the capped simplex.

The call $\mathrm{GMV}(\Sigma, u)$ means Algorithm 2 with zero trading penalty and an equal-weight initial portfolio. Its iteration limit is 5,000 and its stopping criterion is a maximum absolute weight change below 10^{-10}. The common $\mathrm{Validate}$ operation checks dimension N, finite weights, budget error at most 10^{-6}, a minimum weight of at least -10^{-7}, and a maximum weight no greater than $u + 10^{-6}$. For an accepted vector only, it removes negative roundoff by $w_i \leftarrow \max(w_i, 0)$ and normalizes $w \leftarrow w/(\mathbf{1}'w)$ before returning weights aligned to I_t. These checks establish feasibility. Independent optimality checks are reported in Section III-F and the numerical-verification appendix. The procedures describe the implemented allocation rules, conditional on supplied constituents; they do not reconstruct the upstream stock-selection process.

1) Market-Capitalization Weighting with a Position Limit: Market-capitalization weighting assigns capital in proportion to the selected stocks' market values. Brandt, Santa-Clara, and Valkanov [11] use a value-weighted portfolio as a benchmark for assessing active allocation. That literature supports the comparison convention, not a claim that the present 30-stock portfolio or its position limit is optimal. This is a simple allocation benchmark requiring no estimated return or covariance objective.

For formation-date market capitalizations $M_{i,t} > 0$, the uncapped weights are $M_{i,t}/\sum_{j\in I_t} M_{j,t}$. Applying the common position limit while preserving capitalization ratios among

unconstrained stocks gives

$$w_{i,t} = \min\{u, a_t M_{i,t}\}, \qquad \sum_{i \in I_t} w_{i,t} = 1, \tag{6}$$

where $a_t > 0$ redistributes the residual capital and $u = 2/30$. Thus an oversized position is limited and the remainder is allocated proportionally among stocks still below the limit; the process repeats if another stock reaches the limit. MSCI describes this proportional redistribution in its index methodology [12], an industry implementation document, not an academic justification for the choice $2/N$. Its free-float index construction is not replicated here. The 6.67% limit is a study-specific concentration constraint shared with the other allocations; neither cited source establishes this value as optimal or guarantees improved returns.

For historical replication, the backtest uses the selection file's recorded weights, with a Euclidean projection only to correct six-decimal rounding. This rounding correction is distinct from applying a Euclidean projection directly to uncapped capitalization shares, which generally does not preserve their ratios. The team memorandum [10] documents the historical strategy and is used for data provenance, not as academic evidence for weighting. The benchmark averages 19.8 effective holdings and 13.88% monthly turnover, with CAGR 14.05% and Sharpe 1.110 at the baseline cost. These outcomes combine selection and weighting; they do not identify a capitalization premium.

Allocation procedure 1: Market-cap weight

Input: Recorded weights q_t, constituent identifiers I_t, and cap u.

1. Align q_t to the ordered identifiers in I_t. Require finite recorded weights and a total satisfying $|\mathbf{1}'q_t - 1| \le 2 \times 10^{-5}$.
2. Find τ such that $\sum_i [q_{i,t} - \tau]_0^u = 1$.
3. Set $w_{i,t} \leftarrow [q_{i,t} - \tau]_0^u$. This reproduces the recorded allocation up to rounding; no capitalization estimate enters this step.
4. Return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

2) Equal Weighting: Equal weighting assigns identical capital shares without estimating allocation inputs. DeMiguel, Garlappi, and Uppal [2] motivate this benchmark by showing that estimation losses can exceed optimized diversification gains. The rule maximizes effective holdings within the selected set, not variance reduction. Equal capital need not imply equal risk contributions when volatilities or correlations differ. Security selection itself remains estimated.

Each target weight is $1/30$, below the common cap, preserving the unweighted average of selected characteristics. Replacing constituents and reversing price drift still incurs costs. Monthly turnover is 13.62% and Sharpe is 1.087. Annualized mean net return trails market-capitalization weighting by 0.49 percentage points, without significance after multiple-comparison adjustment.

Allocation procedure 2: Equal weighting

Input: Constituent identifiers I_t and cap u.

1. Set $N \leftarrow |I_t|$; require $Nu \ge 1$.
2. For each $i = 1, \ldots, N$, set $w_{i,t} \leftarrow 1/N$.
3. Return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

3) Sample GMVP: Sample GMVP solves the minimum-variance problem (5) with sample covariance and no mean forecast [1]. Unlike marginal-risk rules, it can favor a volatile stock whose covariance with other holdings is low. Avoiding mean estimation removes one source of uncertainty, but estimating 465 covariance entries from 36 observations leaves substantial scope for unstable apparent diversification gains.

Position constraints limit extreme weights without eliminating estimation error or turnover. Average fitted annualized volatility is 8.21%, compared with realized volatility of 10.91%. This contrast is consistent with estimation and distributional instability, although conditional fitted and unconditional realized volatility do not constitute a formal calibration test. Sample GMVP provides the reference for covariance regularization.

Allocation procedure 3: Sample GMVP

Input: Historical return matrix X_t and cap u.

1. Compute $\bar{x} \leftarrow X_t'\mathbf{1}/T$ and $Y \leftarrow X_t - \mathbf{1}\bar{x}'$.
2. Set $S \leftarrow Y'Y/(T-1)$.
3. Set $w_t \leftarrow \mathrm{GMV}(S, u)$; return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

4) Constant-Correlation GMVP: The model preserves individual variances and replaces pairwise correlations with their common average [3], [13]. It minimizes portfolio variance rather than applying inverse-volatility weights. The restriction reduces estimation demands but discards persistent sector and stock-pair dependence. Its usefulness depends on whether this reduction in estimation variance outweighs misspecification and whether marginal risks remain informative.

This trade-off matters in a narrow low-risk universe with limited observations. Mean pairwise correlation is 0.303 and cross-pair dispersion averages 0.176, indicating material heterogeneity. Relative to market-capitalization weighting, the allocation increases the low-volatility percentile by 2.04 percentage points and reduces the negative-coskewness percentile by 1.06 points. Sharpe falls from 1.110 to 1.069; the stronger volatility tilt does not deliver superior net performance.

Allocation procedure 4: Constant-correlation GMVP

Input: Sample covariance S and cap u.

1. Set $\sigma_i \leftarrow \sqrt{S_{ii}}$ and $C_{ij} \leftarrow S_{ij}/(\sigma_i\sigma_j)$.
2. Compute $\bar{\rho} \leftarrow \sum_{i \ne j} C_{ij}/[N(N-1)]$.
3. Set $F_{ii} \leftarrow S_{ii}$ and $F_{ij} \leftarrow \bar{\rho}\sigma_i\sigma_j$ for $i \ne j$.
4. Set $w_t \leftarrow \mathrm{GMV}(F, u)$; return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

5) Ledoit–Wolf Constant-Correlation GMVP: Ledoit–Wolf shrinkage estimates a combination of sample covariance and a constant-correlation target, accounting for their common sampling error [3]. Zero and full shrinkage recover sample and constant-correlation GMVP, respectively, up to covariance scale. Both preserve marginal variances; shrinkage determines how much correlation heterogeneity remains. The derivation assumes temporal independence, finite fourth moments, and the asymptotic conditions in Section III-D, which need not hold in a short financial sample.

Preserving marginal risks allows further defensive allocation, while partial shrinkage can retain sector dependence lost

under the pure target. It can also retain noise that induces trading. Effective holdings average 17.0 versus 16.9 for constant correlation, but turnover rises from 14.41% to 15.65%. MDD is -29.29%, the least severe among covariance GMVPs, yet worse than market-capitalization weighting's -29.00%.

Allocation procedure 5: LW constant-correlation GMVP

Input: Centered returns Y and cap u; noncoincident sample and target matrices.

1. Set $S_0 \leftarrow Y'Y/T$, $v_i \leftarrow (S_0)_{ii}$, and $\sigma_i \leftarrow \sqrt{v_i}$. Construct F_0 by steps 1–3 of Procedure 4, using S_0 in place of S.
2. For each pair (i, j), compute $P_{ij} \leftarrow T^{-1}\sum_s y_{si}^2 y_{sj}^2 - (S_0)_{ij}^2$ and $\Theta_{ij} \leftarrow T^{-1}\sum_s y_{si}^3 y_{sj} - v_i(S_0)_{ij}$. Set $\Theta_{ii} \leftarrow 0$.
3. Set $\widehat{\pi} \leftarrow \sum_{ij} P_{ij}$ and $\widehat{\rho} \leftarrow \sum_i P_{ii} + \bar{\rho}\sum_{i\neq j}(\sigma_j/\sigma_i)\Theta_{ij}$.
4. Compute $\widehat{\gamma} \leftarrow \|F_0 - S_0\|_F^2$ and $\delta \leftarrow [(\widehat{\pi} - \widehat{\rho})/(T\widehat{\gamma})]_0^1$.
5. Set $\widehat{\Sigma} \leftarrow \delta F_0 + (1-\delta)S_0$ and $w_t \leftarrow \mathrm{GMV}(\widehat{\Sigma}, u)$. Return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

6) Ledoit–Wolf Identity GMVP: Ledoit–Wolf identity shrinkage combines sample covariance with a scaled identity target to reduce quadratic estimation loss [14]. Raising small eigenvalues limits the optimizer's reliance on apparently low-risk directions. Unlike constant-correlation shrinkage, it changes variances as well as covariances. Full shrinkage implies equal variances, zero covariances, and equal GMVP weights. Its intensity is estimated, whereas the project's ridge coefficient is fixed.

The target improves conditioning but may suppress genuine volatility differences among selected stocks. Effective holdings rise to 18.2 from sample GMVP's 16.3, while annual cost drag exceeds market-capitalization weighting by 0.14 percentage points. Sharpe is the highest at 1.114 versus 1.110, but CAGR is lower, 13.48% versus 14.05%. This small Sharpe advantage is not statistically significant and does not validate the estimator's loss criterion.

Allocation procedure 6: LW identity GMVP

Input: Centered returns Y and cap u.

1. Set $S_0 \leftarrow Y'Y/T$, $m \leftarrow \mathrm{tr}(S_0)/N$, and $d^2 \leftarrow \|S_0 - mI\|_F^2$.
2. Compute $\bar{b}^2 \leftarrow T^{-2}\sum_s \|y_s y_s' - S_0\|_F^2$.
3. If $d^2 = 0$, set $\delta \leftarrow 1$; otherwise set $\delta \leftarrow \min\{\bar{b}^2, d^2\}/d^2$.
4. Set $\widehat{\Sigma} \leftarrow (1-\delta)S_0 + \delta mI$.
5. Set $w_t \leftarrow \mathrm{GMV}(\widehat{\Sigma}, u)$; return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

7) Ridge GMVP: Ridge GMVP adds a scaled identity matrix to sample covariance, equivalently penalizing squared portfolio weights [14], [15]. The coefficient 0.3 in (14) gives GMVP weights equivalent to fixed identity shrinkage of $0.3/1.3$. This project choice differs from estimated shrinkage and direct inverse-covariance regularization [16]. It raises small eigenvalues and discourages concentration, without penalizing changes from existing holdings.

The fixed penalty balances estimated diversification against weight concentration. Its adequacy can vary across covariance spectra and regimes. Average effective holdings increase from 16.3 under sample GMVP to 17.1, but turnover remains 20.14%. A concentration penalty therefore need not reduce trading. These results concern the specified coefficient, not independently tuned or adaptive ridge allocation.

Allocation procedure 7: Ridge GMVP

Input: Sample covariance S, cap u, and fixed coefficient $\kappa = 0.3$.

1. Compute $m \leftarrow \mathrm{tr}(S)/N$.
2. Set $\widehat{\Sigma} \leftarrow S + \kappa mI$. Do not estimate κ from realized holding-period returns.
3. Set $w_t \leftarrow \mathrm{GMV}(\widehat{\Sigma}, u)$; return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

8) RMT-Clipping GMVP: Random-matrix theory supplies a benchmark for correlation eigenvalues estimated from short samples [17], [18]. Our filter averages eigenvalues below a noise boundary, reconstructs and renormalizes correlation, then solves GMVP. It preserves marginal variances, unlike identity shrinkage, and uses a discrete threshold rather than QIS's nonlinear corrections. Unit noise variance, the N/T boundary, and diagonal normalization are implementation choices, not evidence that all removed spectral differences are noise.

Pooling weak spectral components may stabilize estimated sector relationships or erase genuine dependence. Temporal dependence, changing volatility, and small samples challenge the reference noise model. Gross annual mean return trails market-capitalization weighting by 1.28 percentage points, with 0.12 points of additional costs. This shortfall does not isolate forecast quality; assessing the filter statistically would require covariance-forecast comparisons.

Allocation procedure 8: RMT-clipping GMVP

Input: Sample covariance S, sample length T, and cap u.

1. Set $\sigma_i \leftarrow \sqrt{S_{ii}}$ and $C_{ij} \leftarrow S_{ij}/(\sigma_i\sigma_j)$. Diagonalize $C = U\,\mathrm{diag}(\lambda)U'$.
2. Set $\lambda_+ \leftarrow (1 + \sqrt{N/T})^2$ and $J \leftarrow \{j : \lambda_j < \lambda_+\}$.
3. Initialize $\widetilde{\lambda} \leftarrow \lambda$. If $J \neq \varnothing$, set $\widetilde{\lambda}_j \leftarrow |J|^{-1}\sum_{k\in J}\lambda_k$ for every $j \in J$.
4. Form $A \leftarrow U\,\mathrm{diag}(\widetilde{\lambda})U'$ and $h_i \leftarrow \sqrt{A_{ii}}$.
5. Set $\widehat{\Sigma}_{ij} \leftarrow A_{ij}\sigma_i\sigma_j/(h_i h_j)$. This restores marginal variances after correlation filtering.
6. Set $w_t \leftarrow \mathrm{GMV}(\widehat{\Sigma}, u)$; return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

9) Quadratic Inverse Shrinkage (QIS) GMVP: QIS retains sample covariance eigenvectors and adjusts eigenvalues using inverse eigenvalues and local spectral estimates [4]. Unlike linear shrinkage or threshold clipping, it applies direction-specific nonlinear corrections across the spectrum. It leaves eigenvector error unresolved and estimates covariance, not expected returns. Section III-D states its asymptotic moment, innovation, and spectrum assumptions; supplied author code provides a numerical reference [19].

With 30 stocks and 35 centered degrees of freedom, spectral regularization is relevant, but its estimation criterion omits characteristic exposure and trading costs. QIS averages 17.4 effective holdings and 20.60% monthly turnover. Thus, a richer estimator still produces concentrated, actively traded portfolios. Agreement with author code establishes implementation fidelity, not superior net returns or eigenvector error as the cause of underperformance.

Allocation procedure 9: QIS GMVP

Input: Sample covariance S, sample length T, and cap u.

1. Set $n \leftarrow T-1$, $c \leftarrow N/n$, and $k \leftarrow \min(N,n)$. Diagonalize $S = U\,\mathrm{diag}(\lambda)U'$ in ascending eigenvalue order; remove negative numerical roundoff by $\lambda_j \leftarrow \max(\lambda_j, 0)$.
2. For $j = 1,\ldots,k$, set $a_j \leftarrow 1/\max(\lambda_{N-k+j}, 10^{-12})$. Set $h \leftarrow \min(c^2, c^{-2})^{0.35} N^{-0.35}$.
3. For every (i,j), set $K_{ij} \leftarrow (a_i - a_j)^2 + h^2 a_i^2$. For each j, compute $\theta_j \leftarrow k^{-1}\sum_i a_i(a_i - a_j)/K_{ij}$ and $\psi_j \leftarrow k^{-1}\sum_i h a_i^2/K_{ij}$.
4. If $N \le n$, set $d_j \leftarrow \{a_j[(1-c)^2 + 2c(1-c)\theta_j + c^2(\theta_j^2 + \psi_j^2)]\}^{-1}$.
5. Otherwise, assign $d_{N-k+j} \leftarrow [a_j(\theta_j^2 + \psi_j^2)]^{-1}$ to the nonzero directions and $d_j \leftarrow [(c-1)k^{-1}\sum_i a_i]^{-1}$ to the $N-n$ null directions.
6. Rescale the eigenvalues and reconstruct the covariance:

$$\delta_j \leftarrow d_j \frac{\sum_i \lambda_i}{\sum_i d_i},$$
$$\widehat{\Sigma} \leftarrow U\,\mathrm{diag}(\delta)U'.$$

7. Set $w_t \leftarrow \mathrm{GMV}(\widehat{\Sigma}, u)$; return $\mathrm{Validate}(w_t)$. The current experiment uses $N = 30 < n = 35$.

Output: Target weights w_t for the next holding month.

10) Fama–French Five-Factor (FF5) GMVP: Factor covariance replaces unrestricted dependence with common-factor risk and diagonal residual variance. Extending Sharpe's single-index architecture [20], FF5 GMVP uses market, size, value, profitability, and investment factors [21]. Estimated loadings map factor covariance into stock covariance for variance minimization. This is risk estimation, not factor-premium forecasting. Economic factor definitions distinguish it from spectral methods, while zero residual cross-covariance remains an additional assumption.

Selected low-risk stocks can retain common exposures, but estimation requires stable loadings, adequate factor coverage, and limited residual dependence. Each 36-month regression has only 30 residual degrees of freedom. Average fitted market beta is 0.42 and sample-implied annualized volatility is 8.66%. These diagnostics show reduced estimated risk, without establishing realized protection or preservation of coskewness exposure.

Allocation procedure 10: FF5 GMVP

Input: X_t, aligned factor matrix $F \in \mathbb{R}^{T\times 5}$, historical risk-free returns r_f, and cap u.

1. Form excess returns $X^e \leftarrow X_t - r_f \mathbf{1}_N'$ and the regression design $Z \leftarrow [\mathbf{1}_T, F]$.
2. For each stock i, solve $\widehat{\beta}_i \leftarrow \arg\min_\beta \|X_i^e - Z\beta\|_2^2$ by least squares. Store the five slope coefficients in row i of B; retain the intercept in fitted returns.
3. Set $e_i \leftarrow X_i^e - Z\widehat{\beta}_i$ and $v_i^e \leftarrow e_i'e_i/(T-6)$.
4. Compute $S_f \leftarrow (F - \mathbf{1}\bar{f}')'(F - \mathbf{1}\bar{f}')/(T-1)$. Set $\widehat{\Sigma} \leftarrow BS_fB' + \mathrm{diag}(v^e)$.
5. Set $w_t \leftarrow \mathrm{GMV}(\widehat{\Sigma}, u)$; return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

11) Inverse-Volatility Weighting: Inverse-volatility weighting is the $\eta = 1/2$ variance-power rule [22]: uncapped weights are proportional to reciprocal standard deviation. Estimating marginal variances alone reduces input uncertainty and excludes correlation-driven reallocation. The rule equalizes $w_i\sigma_i$, not risk contributions generally. The two coincide under common correlation before additional constraints; heterogeneous dependence breaks this equivalence.

We impose the cap by projecting normalized inverse-volatility weights onto (4). The square root moderates sensitivity to small estimated variances relative to inverse variance. Effective holdings average 28.7 and Sharpe is 1.086, consistent with a mild additional risk tilt. The rule can miss hedging relationships or overweight temporarily stable stocks. It remains fully invested rather than targeting aggregate volatility through cash or leverage.

Allocation procedure 11: Inverse volatility

Input: Sample covariance S and cap u.

1. For each stock set $s_i \leftarrow \sqrt{\max(S_{ii}, 10^{-14})}$ and $a_i \leftarrow 1/s_i$.
2. Normalize $q_i \leftarrow a_i/\sum_j a_j$.
3. Find τ satisfying $\sum_i [q_i - \tau]_0^u = 1$ and set $w_{i,t} \leftarrow [q_i - \tau]_0^u$.
4. Return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

12) Inverse-Variance Weighting: Inverse variance is the $\eta = 1$ rule of Kirby and Ostdiek [22], with uncapped weights proportional to $1/S_{ii}$. It solves unconstrained fully invested minimum variance under diagonal positive covariance. This analytical equivalence motivates the rule without establishing stock independence. Relative to inverse volatility, it emphasizes low-risk stocks more strongly and is more sensitive to errors in small variance estimates.

Euclidean projection enforces the cap but generally differs from the solution of a bounded diagonal-covariance quadratic program. The policy is cross-sectional allocation, not market-volatility targeting. Its low-volatility percentile exceeds market-capitalization weighting by 0.67 percentage points and Sharpe is 1.083. Correlation omission, volatility shifts, and binding caps limit the interpretation of this stronger defensive tilt.

Allocation procedure 12: Inverse variance

Input: Sample covariance S, cap u, and exponent $\eta = 1$.

1. Set $\ell_i \leftarrow -\eta \log\max(S_{ii}, 10^{-300})$.
2. Set $a_i \leftarrow \exp(\ell_i - \max_j \ell_j)$ and $q_i \leftarrow a_i/\sum_j a_j$. This is numerically stable normalization of inverse variances.
3. Set $w_t \leftarrow \Pi_{\mathcal{W}_u}(q)$ by solving the same scalar budget equation as in Procedure 11.
4. Return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

13) Constrained Risk Budgeting: Risk budgeting distributes portfolio risk rather than capital. Classical equal risk contribution (ERC) equalizes volatility contributions while accounting for covariance [23]; it does not minimize attainable variance. Inverse volatility is a special case under common correlation. Coordinate algorithms [24] support computation, and Richard and Roncalli [25] incorporate investment constraints. Our equal-budget formulation includes bounds directly in (23); binding multipliers can prevent exact ERC.

Positive budgets and stable marginal-risk estimates are required. Risk balancing can sustain broad holdings but does not insure against common shocks. Effective holdings average 27.4, and normalized risk-contribution dispersion is 0.0008 versus 0.0108 under equal weighting. Nevertheless, Sharpe

is lower, 1.072 versus 1.087. Low dispersion neither implies exact constrained ERC nor establishes an economic gain; the adjusted mean-return comparison with market-capitalization weighting is insignificant.

Allocation procedure 13: Constrained risk budgeting

Input: Sample covariance S, equal budgets $b_i = 1/N$, and cap u.

1. Scale $Q \leftarrow S/[\mathrm{tr}(S)/N]$. Initialize a bracket $[a_L, a_H]$ with $a_L = 0$ and $a_H = 2\max\{1, \max_i u^2(Q\mathbf{1})_i/b_i\}$. Initialize $x \leftarrow \mathbf{1}/N$.
2. Set $a \leftarrow (a_L + a_H)/2$ and solve the box-constrained inner problem by cyclic coordinate updates, retaining the previous trial's x as the initial vector.
3. At coordinate i, compute $q_i \leftarrow \sum_{j \neq i} Q_{ij}x_j$ and $d_i \leftarrow Q_{ii}$. Set $x_i \leftarrow \min\{u, [-q_i + \sqrt{q_i^2 + 4d_i a b_i}]/(2d_i)\}$. If $q_i \geq 0$, evaluate the positive root as $2ab_i/[q_i + \sqrt{q_i^2 + 4d_i a b_i}]$.
4. After each sweep, compute $g \leftarrow Qx - ab/x$ componentwise. Put $J_f = \{i : x_i < u - 10^{-12}\}$ and $J_u = J_f^c$. Compute $e_f \leftarrow \max_{i \in J_f} |g_i|$, $e_u \leftarrow \max_{i \in J_u} \max(g_i, 0)$, and $e \leftarrow \max(e_f, e_u)/\max\{1, \|Qx\|_\infty, \|ab/x\|_\infty\}$, taking an empty maximum as zero. Stop the inner iteration when $e \leq 10^{-12}$; raise an error after 2,000 unsuccessful sweeps.
5. If $|\mathbf{1}'x - 1| \leq 10^{-10}$, set $w_t \leftarrow x$ and stop. Otherwise set $a_H \leftarrow a$ when $\mathbf{1}'x > 1$, or $a_L \leftarrow a$ when $\mathbf{1}'x < 1$; return to step 2. Raise an error if 100 trials fail to meet the budget tolerance.
6. Return $\mathrm{Validate}(w_t)$; this last step corrects only budget roundoff. There is no post-solution cap projection.

Output: Target weights w_t for the next holding month.

14) Maximum Diversification: Maximum diversification maximizes weighted standalone volatility relative to portfolio volatility [26]. It rewards covariance offsets and can favor volatile stocks that contribute little joint risk. Unlike ERC, it targets an aggregate ratio rather than prescribed contributions. We solve (26) with sample covariance and common bounds. Its positive numerator permits the convex reformulation (27) and independent optimality checks.

Performance depends on persistent covariance offsets and their return and trading consequences; neither expected returns nor coskewness enters the objective. Effective holdings average 16.4 and turnover is 23.95%. The annual net mean-return shortfall against market-capitalization weighting is 1.53 percentage points: 1.29 before costs and 0.24 from additional costs. Yet realized volatility, 10.71%, is the lowest among all rules, illustrating the distinction between diversification and return maximization.

Allocation procedure 14: Maximum diversification

Input: Sample covariance S and cap u.

1. Set $a_i \leftarrow \sqrt{S_{ii}}$. The numerator $a'w$ is positive for every feasible portfolio.
2. Solve $\min_y y'Sy$ subject to $a'y = 1$, $y \geq 0$, and $y_i \leq u\mathbf{1}'y$, using the positive branch of Algorithm 3.
3. Recover $w_t \leftarrow y/(\mathbf{1}'y)$.
4. Return $\mathrm{Validate}(w_t)$ and record the fitted diversification ratio $a'w_t/\sqrt{w_t'Sw_t}$.

Output: Target weights w_t for the next holding month.

15) Bayes–Stein Sharpe: Jorion's Bayes–Stein estimator shrinks sample means toward a common minimum-variance-weighted mean, with intensity determined by their estimated separation [27]. We combine it with Ledoit–Wolf identity covariance [14] and position bounds to maximize estimated excess return per unit volatility. This hybrid does not reproduce Jorion's predictive covariance or full investment problem. Positive attainable numerators admit (27); uniformly negative numerators require global search.

The estimator assumes stable multivariate-normal sampling and invertible covariance, with $T > N + 2$ only narrowly satisfied by 36 observations and 30 stocks. Shrinkage reduces noisy dispersion without establishing persistent return differences. Fitted monthly mean is 1.36% versus 1.20% for market-capitalization weighting, yet gross annual mean return trails by 1.54 percentage points and net return by 1.74. The Holm-adjusted p-value, 0.266, does not establish significance at 5%.

Allocation procedure 15: Bayes–Stein Sharpe

Input: X_t, sample covariance S, last observed risk-free return $r_{f,t}$, and cap u; $T > N + 2$.

1. Set $\mu \leftarrow X_t'\mathbf{1}/T$ and $\Omega \leftarrow [(T-1)/(T-N-2)]S$. Solve $\Omega v = \mu$ and $\Omega z = \mathbf{1}$.
2. Set $\mu_0 \leftarrow \mathbf{1}'v/(\mathbf{1}'z)$ and $d \leftarrow \mu - \mu_0\mathbf{1}$. Solve $\Omega h = d$.
3. Compute $\phi \leftarrow (N+2)/(N+2+Td'h)$ and $\widehat{\mu} \leftarrow (1-\phi)\mu + \phi\mu_0\mathbf{1}$.
4. Set $a \leftarrow \widehat{\mu} - r_{f,t}\mathbf{1}$. Estimate $\Sigma \leftarrow \widehat{\Sigma}_{LW,I}$ by steps 1–4 of Procedure 6.
5. Obtain $\bar{w} \in \arg\max_{w \in \mathcal{W}_u} a'w$ by filling positions in descending order of a_i; set $m \leftarrow a'\bar{w}$.
6. If $m > 0$, solve (27) with (a, Σ) and set $w_t \leftarrow y/(\mathbf{1}'y)$. If $m = 0$, set $w_t \leftarrow \bar{w}$.
7. If $m < 0$, use the certified vertex search in Algorithm 3 and Appendix B, with relative tolerance 10^{-10}. A node-limit failure must raise an error.
8. Return $\mathrm{Validate}(w_t)$ and retain the sign branch in the solver diagnostics.

Output: Target weights w_t for the next holding month.

16) Minimum CVaR: CVaR measures mean loss in the distribution's upper tail. Rockafellar and Uryasev [28] express empirical minimization as a linear program with a threshold and excess-loss variables, as in (30). It accommodates asymmetric scenarios without assuming Gaussian returns. Unlike variance, it targets losses; unlike drawdown, it measures a single-period tail rather than a path from peak wealth. We use monthly 90% CVaR under the common constraints.

A 36-month window places only 3.6 observations in the 10% empirical tail, including fractional boundary mass. Numerical precision cannot resolve this limited scenario information. The rule does not preserve coskewness exposure explicitly. It averages 15.6 effective holdings and 13.2 capped positions, with MDD -31.57% versus market-capitalization weighting's -29.00%. Minimizing estimated monthly tail loss thus does not lower realized multi-month drawdown here.

Allocation procedure 16: Minimum CVaR

Input: Historical scenarios X_t, tail probability $\alpha = 0.10$, and cap u.

1. Create variables $w \in \mathbb{R}^N$, $\zeta \in \mathbb{R}$, and $z \in \mathbb{R}^T$. Set objective coefficients to $(0_N, 1, \mathbf{1}_T/(\alpha T))$.
2. Impose $\mathbf{1}'w = 1$, $0 \le w_i \le u$, and $z_s \ge 0$. For each historical row x_s' impose $-x_s'w - \zeta - z_s \le 0$.
3. Solve this linear program using Algorithm 6. Leave ζ unrestricted and retain $\alpha T = 3.6$ without integer rounding.
4. On unsuccessful solver termination, raise an error. Otherwise extract w_t and return $\mathrm{Validate}(w_t)$.

Output: Target weights w_t for the next holding month.

17) The GMVP–$1/N$ Mixture: The GMVP–$1/N$ mixture follows the combination principle of Tu and Zhou [29] but fixes the coefficient at one half rather than estimating their optimal mixture. It averages portfolio weights, not covariance matrices, and differs from a ridge penalty. Convexity of (4) ensures feasibility. Equal weighting gives every selected stock positive weight while attenuating sample-GMVP deviations.

The mixture preserves participation in all selected stocks while retaining a covariance tilt. Effective holdings rise from 16.3 to 24.8, without equalizing risk or preserving each characteristic score. The fixed coefficient avoids further estimation but may dilute useful signals or retain noise. Turnover and net returns are not generally averages of the component strategies: they depend on the mixture's own drifting holdings.

Allocation procedure 17: GMVP–$1/N$ mixture

Input: Sample covariance S, cap u, and fixed mixing coefficient $\omega = 0.5$.

1. Compute $g \leftarrow \mathrm{GMV}(S, u)$ and $e \leftarrow \mathbf{1}/N$.
2. Set $w_t \leftarrow \omega g + (1-\omega)e$. Convexity preserves the budget and bounds.
3. Return $\mathrm{Validate}(w_t)$. In the subsequent holding period, use the mixture's own pretrade holdings for transaction-cost accounting.

Output: Target weights w_t for the next holding month.

18) Norm-Constrained GMVP: Portfolio-norm restrictions control estimation-sensitive weights [15]. Bounding squared Euclidean norm limits capital concentration, equivalently imposing a minimum effective holding count. Unlike covariance shrinkage, this directly restricts the feasible set; unlike a penalty, it imposes a hard bound. A binding restriction can exclude unstable portfolios, but also useful concentrated allocations, so its effect depends on the other constraints.

Here $N_{eff} \ge 15$ is redundant because (4) implies $\sum_i w_i^2 \le u = 1/15$. The rule therefore equals sample GMVP in objective, weights, and exposures. Identical Sharpe, 1.096, and effective holdings, 16.3, are required by the specification. We retain the row for transparency and verification, not as evidence about binding norm regularization or an independent economic model.

Allocation procedure 18: Norm-constrained GMVP

Input: Sample covariance S, cap $u = 2/N$, and minimum effective holdings $N/2 = 15$.

1. Set the squared-norm limit $\delta \leftarrow 1/(N/2) = u$.
2. Compute $g \leftarrow \mathrm{GMV}(S, u)$.
3. Use $\sum_i g_i^2 \le u \sum_i g_i = u = \delta$ to verify that the norm constraint is redundant under the current cap.
4. Set $w_t \leftarrow g$ and return $\mathrm{Validate}(w_t)$. No additional ridge term is introduced.

Output: Target weights w_t for the next holding month.

19) Transaction-Cost GMVP: Trading costs make existing holdings relevant to allocation [30]. Our one-period objective (32) adds an L_1 penalty on trades from drifted weights to sample variance. It implements a narrower model than Hautsch and Voigt's treatment of uncertainty. Unlike covariance shrinkage, the penalty changes the objective and depends on inherited positions. It discourages trades with insufficient marginal risk benefit, using proportional costs and fixed risk aversion to set its scale.

Reduced trading can save costs or delay useful adjustment; the penalty is not estimated to be optimal. Liquidation of excluded stocks remains chargeable. Turnover is 12.34%, versus 20.80% for sample GMVP. Relative to market-capitalization weighting, the gross annual mean-return shortfall is 0.59 percentage points; 0.04 points of cost savings reduce it to 0.55. Lower trading therefore does not establish a net-return advantage.

Allocation procedure 19: Transaction-cost GMVP

Input: S, current identifiers I_t, this strategy's previous holdings, baseline realized returns, $c = 0.001$, and $\gamma = 5$.

1. Drift previous holdings using the baseline return ledger. Set terminated positions to zero, align to I_t, and assign zero to new constituents; call the resulting vector p_t. At the first date set $p_t \leftarrow 0$.
2. Retain p_t without renormalizing surviving holdings. Cash and liquidations of excluded securities explain why its entries need not sum to one.
3. Set $\lambda \leftarrow 2c/\gamma = 0.0004$.
4. Set $L \leftarrow 2\lambda_{\max}(S) \times 1.0001 + 10^{-18}$. Use $v^0 \leftarrow p_t$ if $\mathbf{1}'p_t > 0$, and $v^0 \leftarrow \mathbf{1}/N$ otherwise. Obtain the feasible initial w^0 by applying (34) to v^0 with this L and p_t.
5. Starting from w^0, use Algorithm 2 to minimize $w'Sw + \lambda\|w - p_t\|_1$ over $\mathcal{W}_u$, applying the same joint proximal map.
6. Return $\mathrm{Validate}(w_t)$. Charge trades in all securities, including mandatory liquidations, in the separate return ledger.

Output: Target weights w_t for the next holding month.

20) Implications of the Cross-Method Comparison: Statistical estimation, numerical optimization, and investment performance involve distinct criteria. Minimizing covariance loss or accurately solving an empirical objective does not guarantee higher future net returns. Constant correlation and QIS change the estimator within a common objective; risk budgeting and CVaR change the objective itself. Realized Sharpe rankings alone cannot identify which statistical assumptions are more accurate.

Reweighting also changes selection exposures. None of the objectives directly maximizes the negative-coskewness score. Minimum variance can strengthen low-volatility exposure, risk budgets distribute estimated risk, and maximum Sharpe introduces mean forecasts. Portfolio-weighted scores and return/cost decompositions assess these effects against each rule's stated criterion.

D. Covariance Estimators

1) Constant Correlation: Let $\sigma_i = \sqrt{S_{ii}}$, $D = \mathrm{diag}(\sigma)$, and

$$\bar{\rho} = \frac{1}{N(N-1)} \sum_{i \ne j} \frac{S_{ij}}{\sigma_i \sigma_j}. \tag{7}$$

The constant-correlation estimator [13] is

$$F = (1-\bar{\rho})D^2 + \bar{\rho}\sigma\sigma'. \tag{8}$$

It preserves each marginal variance while replacing 435 pairwise correlations by their average. For a valid correlation matrix, $-1/(N-1) \le \bar{\rho} \le 1$, so F is positive semidefinite; strict inequalities and positive variances give positive definiteness. The statistical restriction concerns dependence, not equal asset volatility. When unconstrained and nonsingular, its minimum-variance weights are proportional to

$$\sigma_i^{-2} - \frac{\bar{\rho}}{1+(N-1)\bar{\rho}} \left(\sum_j \sigma_j^{-1} \right) \sigma_i^{-1}. \tag{9}$$

This expression explains why the method is neither equal weighting nor inverse-volatility weighting. Estimation variance is reduced at the cost of bias when correlations differ systematically across sectors or stocks.

2) Ledoit–Wolf Constant-Correlation Shrinkage: Here the sample and target matrices are constructed using divisor T. Let $s_{ij} = (S_0)_{ij}$, $v_i = s_{ii}$, and let F_0 be (8) formed from S_0. The estimator [3] is

$$\widehat{\Sigma}_{LW,CC} = \delta F_0 + (1-\delta)S_0, \qquad \delta = \left[\frac{\widehat{\pi} - \widehat{\rho}}{T\widehat{\gamma}} \right]_0^1, \tag{10}$$

where $[z]_0^1 = \min(1, \max(0, z))$ and

$$\begin{aligned} \widehat{\pi}_{ij} &= T^{-1} \sum_s y_{si}^2 y_{sj}^2 - s_{ij}^2, \\ \widehat{\theta}_{ij} &= T^{-1} \sum_s y_{si}^3 y_{sj} - v_i s_{ij}, \\ \widehat{\pi} &= \sum_{ij} \widehat{\pi}_{ij}, \qquad \widehat{\gamma} = \|F_0 - S_0\|_F^2, \\ \widehat{\rho} &= \sum_i \widehat{\pi}_{ii} + \bar{\rho} \sum_{i \ne j} \sqrt{v_j/v_i}\, \widehat{\theta}_{ij}. \end{aligned} \tag{11}$$

The off-diagonal multiplier is the standard-deviation ratio, not its square root. Coincident target and sample matrices give the same estimator at any intensity. We retain the original $1/T$ normalization, distinguishing it from the later author code's default degrees-of-freedom convention. Frobenius-risk improvement does not guarantee higher net portfolio returns.

3) Ledoit–Wolf Identity Shrinkage and Ridge: Define $m = \operatorname{tr}(S_0)/N$, $d^2 = \|S_0 - mI\|_F^2$, and

$$\bar{b}^2 = T^{-2} \sum_s \|y_s y_s' - S_0\|_F^2. \tag{12}$$

With $b^2 = \min(\bar{b}^2, d^2)$, identity shrinkage [14] uses

$$\widehat{\Sigma}_{LW,I} = (1 - b^2/d^2)S_0 + (b^2/d^2)mI, \tag{13}$$

with the coincident-target case treated directly. Its target changes marginal variances as well as covariances. In contrast, the project's ridge rule is

$$\widehat{\Sigma}_{ridge} = S + 0.3\ \operatorname{tr}(S)I/N. \tag{14}$$

For GMVP weights, (14) is equivalent to identity shrinkage with fixed intensity $0.3/1.3$, or to adding $0.3\operatorname{tr}(S)\|w\|_2^2/N$ to the sample-variance objective. The value 0.3 is a project specification, not an estimated optimal intensity. This operation is distinct from directly shrinking an inverse covariance matrix [16].

4) Random-Matrix Correlation Filtering: Decompose $C = D^{-1}SD^{-1} = U\operatorname{diag}(\lambda)U'$. The implemented filter uses the independent, unit-variance noise boundary

$$\lambda_+ = (1 + \sqrt{N/T})^2. \tag{15}$$

Every eigenvalue below λ_+ is replaced by the mean of those eigenvalues; larger eigenvalues are retained. Let $A = U\operatorname{diag}(\lambda)U'$ and $H = \operatorname{diag}(\sqrt{A_{11}}, \ldots, \sqrt{A_{NN}})$. The final estimate is

$$\widehat{\Sigma}_{RMT} = DH^{-1}AH^{-1}D. \tag{16}$$

Pooling preserves trace before diagonal renormalization, which restores unit correlation diagonal but generally changes the eigenvalues again. Unit noise variance and the use of T are project conventions motivated by [17], [18]. Financial dependence and finite samples limit the noise benchmark; eigenvalues below its boundary need not be uninformative.

5) Quadratic Inverse Shrinkage: QIS [4] keeps sample eigenvectors and estimates nonlinear eigenvalue corrections. Put $n = T-1$, $c = N/n$, $S = U\operatorname{diag}(\lambda_1, \ldots, \lambda_N)U'$ with ascending eigenvalues, and $k = \min(N, n)$. For the k nonzero eigenvalues write $a_j = 1/\lambda_{N-k+j}$ and

$$\begin{aligned} h &= \min(c^2, c^{-2})^{0.35} N^{-0.35}, \\ \theta_j &= \frac{1}{k} \sum_{i=1}^{k} \frac{a_i(a_i - a_j)}{(a_i - a_j)^2 + h^2 a_i^2}, \\ \psi_j &= \frac{1}{k} \sum_{i=1}^{k} \frac{h a_i^2}{(a_i - a_j)^2 + h^2 a_i^2}. \end{aligned} \tag{17}$$

For $N \le n$, the preliminary eigenvalue estimate is

$$d_j = \{a_j[(1-c)^2 + 2c(1-c)\theta_j + c^2(\theta_j^2 + \psi_j^2)]\}^{-1}. \tag{18}$$

For $N > n$, nonzero directions use $d_j = [a_j(\theta_j^2 + \psi_j^2)]^{-1}$ and the $N-n$ null directions share $d_0 = [(c-1)k^{-1}\sum_i a_i]^{-1}$. Finally,

$$\delta_j = d_j \frac{\sum_i \lambda_i}{\sum_i d_i}, \qquad \widehat{\Sigma}_{QIS} = U\operatorname{diag}(\delta)U'. \tag{19}$$

The experiment has $30 < 35$, but both dimensional branches are tested. Eigenvalues are floored at 10^{-12} before inversion. The estimator follows [4, eqs. (4.12)–(4.15), (6.3)] and is checked against the supplied January 2024 author code [19]. Its rotation-equivariant asymptotics require a stable population spectrum bounded away from zero and infinity, an aspect ratio bounded away from zero and one, and standardized independent innovations with a finite 16th moment under the stated theorems. Suggested weaker moment conditions are conjectural. These results do not guarantee superior GMVP returns at $T = 36$.

6) Five-Factor Covariance: For each stock estimate by ordinary least squares

$$r_{i,s} - r_{f,s} = \alpha_i + \beta_i' f_s + \varepsilon_{i,s}, \tag{20}$$

where f_s contains Mkt–RF, SMB, HML, RMW, and CMA [21]. With $K_f = 5$, the estimate is

$$\widehat{\Sigma}_F = \widehat{B} S_f \widehat{B}' + \operatorname{diag}\left(\frac{\sum_s \widehat{\varepsilon}_{i,s}^2}{T - K_f - 1} \right). \tag{21}$$

Regressions include an intercept and aligned dates; factor covariance uses divisor $T - 1$. The single-index extension [20] assumes zero residual cross-covariances. Excess-return covariance treats the next holding-period risk-free return as predetermined. The five-factor paper defines the factors rather than establishing this estimator's optimality. Only 30 residual degrees of freedom remain.

E. Alternative Allocation Objectives

1) Inverse Volatility and Inverse Variance: Given marginal sample variances, define

$$q_i(\eta) = \frac{(S_{ii})^{-\eta}}{\sum_j (S_{jj})^{-\eta}}, \qquad w = \Pi_{\mathcal{W}_u}(q(\eta)). \tag{22}$$

Inverse volatility uses $\eta = 1/2$; inverse variance uses $\eta = 1$ [22]. Projection is an added project constraint. Both rules ignore pairwise correlations, are scale invariant, and remain fully invested rather than controlling portfolio volatility through cash or leverage.

2) Box-Constrained Risk Budgeting: For $b_i = 1/N$ and positive-definite S, unconstrained ERC satisfies $w_i(Sw)_i = b_i w' S w$ [23], [24]. We impose bounds directly through Richard and Roncalli's formulation [25, eq. (11)], rather than projecting an unconstrained ERC solution.

For $a > 0$, solve the strictly convex problem

$$x(a) = \arg \min_{0 < x_i \le u} \left\{ \tfrac{1}{2} x' S x - a \sum_i b_i \log x_i \right\}. \tag{23}$$

Choose a so that $\mathbf{1}' x(a) = 1$ and set $w = x(a)$, without subsequent constraint projection. The common output adapter removes only the remaining budget roundoff (below 10^{-10} in the inner solver). This variance formulation has the same stationarity conditions as the volatility formulation $\sqrt{x'Sx} - \lambda \sum_i b_i \log x_i$ at $\lambda = a/\sqrt{x'Sx}$. At an uncapped coordinate, $x_i(Sx)_i = a b_i$; at an upper bound,

$$x_i(Sx)_i = a b_i - \nu_i x_i, \quad \nu_i \ge 0, \quad \nu_i(x_i - u) = 0. \tag{24}$$

Consequently, binding constraints generally prevent equal final risk contributions, as described in [25, eq. (17)]. We therefore label this rule *constrained risk budgeting*, while retaining the identifier `erc` solely for continuity of replication files.

Fixing the other coordinates, let $q_i = \sum_{j \ne i} S_{ij} x_j$. The exact positive coordinate solution before enforcing the bound is

$$x_i^+ = \frac{-q_i + \sqrt{q_i^2 + 4 S_{ii} a b_i}}{2 S_{ii}}. \tag{25}$$

The update is $x_i \leftarrow \min(u, x_i^+)$; a rationalized expression avoids cancellation when $q_i > 0$. Inner convergence is checked using box stationarity, and outer bracketing adjusts a to meet the budget. This supplies an economically defined constrained model rather than a repair of weights after optimization.

3) Maximum Diversification and a Convex Reformulation: The maximum-diversification portfolio [26] solves

$$\max_{w \in \mathcal{W}_u} \frac{\sigma' w}{\sqrt{w' S w}}. \tag{26}$$

The numerator measures weighted standalone volatility and the denominator joint volatility. Favorable covariance can justify a large weight in a volatile stock; expected returns and original characteristic exposures are not targeted.

More generally, for a vector a admitting $a'w > 0$, define $y = w/(a'w)$. The positive-numerator ratio problem is equivalent to

$$\begin{aligned} &\min_y \ y' \Sigma y, \\ &\text{subject to } a' y = 1, \quad y_i \ge 0, \quad y_i \le u \mathbf{1}' y. \end{aligned} \tag{27}$$

Conversely, feasible y yields $w = y/(\mathbf{1}'y) \in \mathcal{W}_u$ with ratio $(y' \Sigma y)^{-1/2}$, proving equivalence. For $a = \sigma > 0$, every feasible numerator is positive. We solve this quadratic program with feasibility and stationarity checks. The reformulation replaces fixed-step ascent and is derived here, not attributed to the empirical source paper.

4) Bayes–Stein Mean and Maximum Estimated Sharpe Ratio: Let $\mu = \bar{x}$ and $\Omega = [(T-1)/(T-N-2)]S$. The implemented Bayes–Stein estimate [27] is

$$\begin{aligned} \mu_0 &= \frac{\mathbf{1}' \Omega^{-1} \mu}{\mathbf{1}' \Omega^{-1} \mathbf{1}}, \\ \phi &= \frac{N+2}{N + 2 + T(\mu - \mu_0 \mathbf{1})' \Omega^{-1} (\mu - \mu_0 \mathbf{1})}, \\ \widehat{\mu}_{BS} &= (1 - \phi)\mu + \phi \mu_0 \mathbf{1}. \end{aligned} \tag{28}$$

The correction requires $T > N + 2$ and nonsingular covariance; its multiplier here is $35/4$. The derivation assumes independent multivariate-normal returns with stable means and positive-definite covariance. These sampling conditions are not established by the backtest.

We use $a = \widehat{\mu}_{BS} - r_{f,t}\mathbf{1}$ and $\Sigma = \widehat{\Sigma}_{LW,I}$ in (26). The last observed risk-free rate is a forecast convention. Combining this mean with identity shrinkage and position bounds defines a hybrid rather than Jorion's complete Bayesian portfolio problem.

We first maximize the linear numerator over $\mathcal{W}_u$ by filling positions in descending order of a_i. A positive maximum permits (27). If the maximum is zero, the corresponding zero-numerator portfolio is optimal. If every feasible numerator is strictly negative, the positive-numerator transformation is invalid. Put $b = -a$, so $b'w > 0$ on $\mathcal{W}_u$. Maximizing the negative ratio is equivalent to maximizing $\sqrt{w'\Sigma w}/(b'w)$. For $w = \sum_v \alpha_v v$, a convex combination of vertices,

$$\frac{\|\Sigma^{1/2} w\|_2}{b'w} \le \sum_v \frac{\alpha_v b' v}{b' w} \frac{\|\Sigma^{1/2} v\|_2}{b' v} \le \max_v \frac{\|\Sigma^{1/2} v\|_2}{b' v}. \tag{29}$$

Thus a global optimum occurs at a vertex. With $u = 1/15$ and $N = 30$, each vertex has 15 weights equal to u and 15 zeros, giving $\binom{30}{15} = 155{,}117{,}520$ candidates. A branch-and-bound search compares this finite vertex set, pruning only when an admissible bound excludes an improvement beyond relative tolerance 10^{-10}. A locally converged ratio solution is not substituted for this certificate.

5) Historical Conditional Value at Risk: For historical scenario losses $\ell_s(w) = -x_s'w$ and tail probability $\alpha = 0.10$, the Rockafellar–Uryasev formulation [28] is

$$\min_{w,\zeta,z} \zeta + \frac{1}{\alpha T}\sum_{s=1}^{T} z_s,$$
$$\text{subject to } w \in \mathcal{W}_u, \quad z_s \geq -x_s'w - \zeta, \quad z_s \geq 0. \tag{30}$$

The threshold ζ is unrestricted; nonnegativity would exclude valid loss distributions. Tail probability 0.10 corresponds to 90% confidence, with fractional mass $\alpha T = 3.6$. Primal and dual feasibility are checked, and solver failure raises an error without substituting another rule. Numerical accuracy does not ensure accurate future-tail estimation.

6) Weight Mixing and the Redundant Norm Constraint: The fixed mixture $w_{mix} = 0.5w_{GMVP} + 0.5\mathbf{1}/N$ is feasible by convexity. It follows the combination principle [29], not that paper's estimated optimal coefficient. A partial Kan–Wang–Zhou code citation remains unresolved provenance [31], rather than a verified derivation of this rule.

Norm restrictions can regularize portfolio weights [15]. In this experiment, however, $N_{eff} \geq 15$ requires $\|w\|_2^2 \leq 1/15$, and

$$\sum_i w_i^2 \leq u \sum_i w_i = u = 1/15 \tag{31}$$

already follows from (4). The implementation returns the same sample GMVP when the norm constraint is inactive; it does not add an unreported small ridge penalty. Identical results in these two rows are therefore required by the model definitions.

7) Transaction-Cost-Penalized Minimum Variance: Let p_t contain pretrade weights after price drift, aligned to the currently eligible securities. The problem is

$$\min_{w\in\mathcal{W}_u} w'Sw + \lambda\|w - p_t\|_1, \qquad \lambda = 2c/\gamma = 0.0004, \tag{32}$$

where $c = 0.001$ is cost per dollar traded and $\gamma = 5$. This scaling follows by minimizing $\gamma w'Sw/2 + c\|w - p_t\|_1$ and multiplying by $2/\gamma$. Liquidating excluded securities adds a constant cost conditional on the selected set, so it affects realized returns but not this optimizer. The relation between costs and regularization is discussed in [30]; our one-period quadratic-plus-L_1 problem does not reproduce that paper's complete treatment of model uncertainty.

The penalty depends on the pretrade portfolio and therefore makes this allocation path dependent. Estimation and accounting use the baseline return scenario to update that state. Cost sensitivities hold the target-weight sequence fixed; they assess accounting sensitivity rather than a separately reoptimized policy at each cost rate.

F. Numerical Algorithms and Verification

For a vector v, the capped-simplex projection is

$$[\Pi_{\mathcal{W}_u}(v)]_i = [v_i - \tau]_0^u, \quad \sum_i [v_i - \tau]_0^u = 1. \tag{33}$$

The scalar equation is monotone and is solved by safeguarded root finding. For the transaction-cost problem, the combined proximal operator is

$$w_i = [p_i + \mathcal{S}_{\lambda/L}(v_i - \tau - p_i)]_0^u, \tag{34}$$

where $\mathcal{S}_a(z) = \text{sign}(z)\max(|z| - a, 0)$ and τ again enforces the budget. Solving for τ jointly with thresholding is essential: separate soft thresholding and simplex projection need not equal the constrained proximal map.

The quadratic and composite problems use accelerated proximal iterations based on FISTA [32], with gradient-based adaptive restart [33]. The smooth gradient is $2\Sigma w$, and L is chosen above $2\lambda_{max}(\Sigma)$. The implementation uses an iterate tolerance of 10^{-10} and at most 5,000 iterations; finite stopping is supplemented by independent optimality diagnostics, rather than described as an exact finite-step solution. For a differentiable convex objective, the Frank–Wolfe gap

$$G(w) = \max_{v\in\mathcal{W}_u} \nabla f(w)'(w - v) \tag{35}$$

satisfies $0 \leq f(w) - f^* \leq G(w)$. The inner linear minimization fills weights in ascending gradient order. Transaction-cost diagnostics additionally use an epigraph representation of absolute deviations; CVaR is checked against its dual linear program. Feasibility alone cannot certify optimality.

Diagnostics include $N_{eff} = 1/\sum_i w_i^2$, active bounds, fitted volatility, normalized risk contributions, and portfolio-weighted scores. A common sample covariance isolates weight differences in cross-model diagnostics. Superseded outputs are archived separately from the corrected performance results.

G. Timing and Costs

Inputs use the 36 months through t and weights earn returns at $t+1$. Execution is assumed at the rebalancing close without an additional lag. Portfolios are fully invested, unlevered, and constrained at rebalancing:

$$\mathbf{1}'w_t = 1, \qquad 0 \leq w_{i,t} \leq 2/30. \tag{36}$$

The recorded benchmark weights are rounded to six decimal places. Projection onto the common feasible set restores the budget and position constraints, with a maximum stock-level adjustment of 3.33×10^{-7}.

Let $\widetilde{w}_t$ denote portfolio weights immediately before rebalancing, after accounting for the preceding holding-period returns and any residual cash. Total traded notional as a fraction of portfolio value, D_t, and the subsequent net return are

$$D_t = \sum_i |w_{i,t} - \widetilde{w}_{i,t}|, \quad R_{t+1}^{net} = (1 - cD_t)(1 + w_t'r_{t+1}) - 1. \tag{37}$$

Traded notional includes purchases and liquidations: $D_t = 1$ for initial investment from cash and $D_t = 2$ for complete replacement. Cost per dollar traded is $c = 0.001$ (10 basis points), equivalent to 20 basis points on reported half-L_1 turnover: $cD_t = 0.002(D_t/2)$. Initial investment costs 10 basis points and complete replacement 20. Termination proceeds remain in cash until rebalancing; no final forced-liquidation charge is imposed.

H. Market Benchmarks

Market benchmarks evaluate the combined selection and weighting strategy; Market-cap weight and equal weighting assess allocation within the same 30 stocks. S&P 500 represents

Algorithm 1: Monthly estimation and allocation

1) Input constituents selected at t, their returns through t, factor observations, pretrade weights, and fixed settings (T, N, u, c, γ).
2) Require 36 complete historical observations for every selected stock; retain the same constituents for every allocation rule.
3) Center the return matrix. Estimate S, S_0, marginal volatilities, and the covariance estimator specified in Section III-D.
4) Solve the stated allocation problem using the corresponding allocation procedure above, Algorithms 2–4 for optimization, Algorithm 5 for covariance estimation, or Algorithm 6 for CVaR. Check budget, bounds, finite weights, and the relevant numerical certificate.
5) Apply the weights to month $t+1$. Calculate total traded notional, multiplicative net returns, and drifted pretrade weights; handle security termination under the stated cash convention.
6) Store inputs, weights, solver diagnostics, and realized returns. Only after the full path is complete, calculate performance and the prespecified regime summaries.

Fig. 2. Investment timing and separation of estimation, optimization, and evaluation.

Algorithm 2: Accelerated constrained quadratic/composite minimization

1) Input Σ, u, previous weights p, and penalty $\lambda \geq 0$. Set $L > 2\lambda_{max}(\Sigma)$, a feasible w^0, $y^0 = w^0$, and $a_0 = 1$.
2) Compute $v^k = y^k - 2\Sigma y^k/L$. Obtain w^{k+1} by (34), solving the budget equation for τ; $\lambda = 0$ gives ordinary projection.
3) If the iterate change meets tolerance, stop iteration and evaluate feasibility and optimality diagnostics.
4) If $(y^k - w^{k+1})'(w^{k+1} - w^k) > 0$, restart with $a_{k+1} = 1$ and $y^{k+1} = w^{k+1}$.
5) Otherwise set $a_{k+1} = (1 + \sqrt{1 + 4a_k^2})/2$ and $y^{k+1} = w^{k+1} + [(a_k - 1)/a_{k+1}](w^{k+1} - w^k)$; repeat from step 2.
6) Report the numerical tolerance and independent gap/residual. Do not infer a global optimum solely from a small change in consecutive iterates.

Fig. 3. FISTA-type implementation with a jointly constrained proximal map.

Algorithm 3: Global ratio optimization by numerator sign

1) Input positive-definite Σ, numerator vector a, and cap u. Calculate $m = \max_{w \in \mathcal{W}_u} a'w$ by a greedy linear allocation.
2) If $m > 0$, solve the convex quadratic program (27); check primal feasibility and stationarity, and return $w = y/(\mathbf{1}'y)$.
3) If $m = 0$, return a feasible linear maximizer, whose ratio is zero.
4) If $m < 0$, search the vertices of $\mathcal{W}_u$ by branch and bound, pruning only with the lower bound in Appendix B. Retain the best vertex with a deterministic tie rule.
5) Record which branch was used. The vertex argument and the admissible pruning bound certify step 4 to the stated relative tolerance; the convex formulation establishes globality of step 2 up to numerical tolerance.

Fig. 4. Sign-aware treatment of maximum diversification and estimated maximum Sharpe.

large-cap U.S. equity performance, while CRSP and FF provide broader market references. The first is a price-return series; the latter two include distributions. All share the same 215 holding months, risk-free series, and performance formulas. No strategy trading cost is deducted from published indices. These comparisons neither implement an index-tracking fund nor isolate a causal characteristic premium.

1) S&P 500 Price Index: The primary market series is CRSP's published S&P 500 Composite return, originally named `sprtrn` and supplied as `sp500_return`. It is calculated from consecutive month-end index levels:

$$R_t^{SP} = I_t^{SP}/I_{t-1}^{SP} - 1. \tag{38}$$

The supplied monthly returns are used directly; they are not summed across days or reconstructed from current constituents. CRSP documents this published series as excluding dividends [34]. It must also be distinguished from CRSP's separately constructed value-weighted portfolio of S&P 500 constituents.

The S&P row has CAGR 8.61%, volatility 14.30%, Sharpe 0.566, and MDD -52.56%. Its bold label and underlined table values, and thick orange graph line, identify the primary benchmark rather than statistical significance. Because strategy returns include distributions, the growth difference is not a dividend-matched excess return or alpha. CRSP and FF retain total-return market comparisons, but neither substitutes for an official S&P 500 Total Return series, which is not present in the supplied data.

2) CRSP Value-Weighted Market: The supplied CRSP value-weighted market series includes distributions [34]. Its capitalization weighting can be written as $R_t^{CRSP} = \sum_{i \in \mathcal{E}_t} v_{i,t-1} R_{i,t}$, where $v_{i,t-1}$ is prior-period market capitalization normalized over the eligible securities $\mathcal{E}_t$. CRSP determines eligibility and corporate-event treatment; we use its published series

Algorithm 4: Risk budgeting with box constraints

1) Input positive-definite S, budgets $b_i > 0$ summing to one, and cap u. Bracket a positive parameter a for which the solution of (23) crosses unit total weight.
2) For a trial a, cycle over coordinates using the positive quadratic root and the upper bound. Use the latest coordinates when evaluating each covariance-row product.
3) Continue until box-constrained stationarity meets tolerance. Check zero gradient on uncapped coordinates and the appropriate gradient sign on capped coordinates.
4) Adjust the bracket according to $\mathbf{1}'x(a) - 1$ and repeat until the budget residual meets tolerance.
5) Return $w = x(a)$ without a final constraint projection; the common adapter removes only budget roundoff. Report risk contributions and active bounds. Do not claim exact equal risk contributions where upper bounds bind.

Fig. 5. Coordinate minimization and budget calibration for constrained risk budgeting.

Algorithm 5: Covariance estimation for the eight GMVP specifications

1) Input the common $T \times N$ return matrix and aligned factors. Calculate Y, S, S_0, and positive marginal standard deviations. Select one of the following branches; do not average across branches.
2) *Sample / constant correlation:* return S, or replace all off-diagonal sample correlations by $\bar{\rho}$ and reconstruct F using (8).
3) *LW constant correlation:* construct F_0 from S_0; calculate the fourth-moment estimates $\widehat{\pi}$, $\widehat{\rho}$, and $\widehat{\gamma}$; truncate the resulting intensity to $[0, 1]$ and return the convex combination.
4) *LW identity / ridge:* for the former, estimate b^2/d^2 from centered observations and combine S_0 with mI; for the latter, add $0.3\,\mathrm{tr}(S)I/N$ to S without estimating an intensity.
5) *RMT:* diagonalize the sample correlation matrix, average eigenvalues below the stated noise boundary, reconstruct and normalize its diagonal, then restore marginal standard deviations.
6) *QIS:* diagonalize S in ascending order; compute inverse eigenvalues, h, θ, and ψ with the specified index orientation; apply the appropriate dimensional branch, normalize the sum of corrected eigenvalues, and reconstruct with the same eigenvectors.
7) *FF5:* regress each stock's excess returns on an intercept and five factors; combine the fitted factor covariance with diagonal residual variances using the residual degrees of freedom.
8) Check finiteness, symmetry, and positive semidefiniteness. Use the resulting matrix in Algorithm 2 with $\lambda = 0$ and the common cap. Retain the estimator identity and parameter settings in the replication record.

Fig. 6. Estimator-specific operations; the statistical formulas and limitations are given in Section III-D.

rather than reconstructing it from the incomplete project panel. This broad-market reference differs from both the S&P 500 constituent universe and the selected 30-stock capped portfolio.

CRSP has CAGR 10.86%, annualized volatility 14.93%, Sharpe 0.687, and MDD -51.48%. Including distributions makes its return convention consistent with the strategy series, although constituent coverage and investment costs still differ.

3) Fama–French Market: The FF market total return is $(\mathrm{Mkt} - \mathrm{RF}) + \mathrm{RF}$ from the monthly factor file [21]. Adding RF aligns its excess-return input with strategy total returns before performance calculations subtract RF. This precomputed market series differs from FF5 GMVP, which uses five factors to estimate selected-stock covariance and optimize weights.

The FF benchmark preserves continuity with the original study and checks sensitivity to the market definition. Its CAGR is 11.54%, volatility 14.83%, Sharpe 0.732, and MDD -50.31%. The three market series are distinct: S&P represents large-cap price growth, while CRSP and FF provide total-return references with their own coverage and data versions. Their differences should not be attributed solely to dividends, and they are not independent tests of a selection premium.

I. Performance Metrics and Statistical Inference

The compound annual growth rate (CAGR) is $[\prod_{t=1}^{T}(1 + R_t^{net})]^{12/T} - 1$. Annualized volatility is the sample standard deviation of monthly net returns multiplied by $\sqrt{12}$. The annualized Sharpe ratio is $\sqrt{12}\,\overline{(R^{net} - R_f)}/s(R^{net} - R_f)$, where R_f is the contemporaneous monthly risk-free return. Maximum drawdown (MDD) is the minimum percentage decline from the preceding wealth maximum, including initial wealth normalized to one. Annualized arithmetic mean return, $12\overline{R}$, is used in the return decomposition. The original research also reports $(1+\overline{R})^{12}-1$; this convention is identified separately because it differs from CAGR.

Mean-return differences are tested through constant-only regressions of monthly paired net returns, using four-lag Bartlett Newey–West standard errors [35]. Two-sided asymptotic tests receive Holm adjustment across 18 comparisons within each benchmark, scenario, and period [36]. Sharpe differences use pointwise intervals from 2,000 paired circular block-bootstrap replications, six-month blocks, and seed 20261002 [37]. Neither procedure adjusts for the original strategy search.

For market comparisons, $\Delta SR = SR_P - SR_M$ compares market-capitalization weighting with the specified market

Algorithm 6: Historical CVaR linear program

1) Input X, tail probability $\alpha \in (0,1)$, and cap u. Introduce variables (w, ζ, z) with objective coefficients $(0_N, 1, \mathbf{1}_T/(\alpha T))$.
2) Set the equality $\mathbf{1}'w = 1$. For each scenario impose $-x_s'w - \zeta - z_s \leq 0$.
3) Set bounds $0 \leq w_i \leq u$, $z_s \geq 0$, and $\zeta \in \mathbb{R}$; do not round αT or restrict the sign of the threshold.
4) Solve the linear program. Require successful termination and feasible primal variables; independently compare its value with the dual optimum.
5) Return the stock weights. If the program fails, stop that experiment with an error instead of replacing the allocation rule.

Fig. 7. Explicit construction of the empirical loss-minimization problem.

on identical months. Each bootstrap draw jointly resamples portfolio, market, and risk-free returns; percentile intervals are the 2.5th and 97.5th percentiles of the resulting differences. Each temporal subsample is resampled separately. Three- and twelve-month blocks assess sensitivity to the baseline six-month choice. The proportion of unrestricted bootstrap differences below zero is reported only as a descriptive tail frequency: the resampling does not impose $H_0 : \Delta SR = 0$. These intervals assume approximate stationarity within each evaluation window and blocks long enough to capture relevant dependence; structural change can impair coverage. They are not corrected for specification search or simultaneous comparisons across markets and periods.

Factor-adjusted performance is estimated separately from the FF5 covariance allocation rule:

$$R_{P,t}^{net} - R_{f,t} = \alpha + \beta' f_t + \varepsilon_t, \tag{39}$$

where f_t contains the FF market excess return, SMB, HML, RMW, and CMA [21]. OLS estimates use the same monthly observations as the corresponding Sharpe comparison. We report $12\widehat{\alpha}$, an arithmetic annualization of the monthly intercept, and $t_{\mathrm{HAC}} = \widehat{\alpha}/\widehat{\mathrm{se}}_{\mathrm{HAC}}(\widehat{\alpha})$. The standard error uses four-lag Bartlett HAC without a degrees-of-freedom multiplier; two-sided p values use the asymptotic standard normal distribution. Alpha measures performance unexplained by these five factors, rather than a benchmark CAGR difference or a model-free measure of skill.

IV. Portfolio Performance

A. Full-Period Performance

Allocation and market labels in Table II link to their methodological reviews and benchmark definitions.

Table II shows no uniformly superior rule. LW identity has the highest Sharpe (1.114), but lower CAGR than market-capitalization weighting (13.48% versus 14.05%; benchmark Sharpe 1.110). Favorable extrema are bold; benchmark return and Sharpe are underlined. Maximum diversification has the lowest volatility (10.71%), Market-cap weight the least severe MDD (-29.00%), and transaction-cost GMVP the lowest turnover (12.34%). The underlined minimum Sharpe is 0.995 for Bayes–Stein.

All 19 strategy CAGRs, 12.15–14.05%, exceed S&P 500's underlined 8.61%, CRSP's 10.86%, and FF's 11.54%. Market-cap weight's advantage over the price index is 5.44 percentage points in CAGR, with MDD -29.00% versus S&P's underlined -52.56%. S&P volatility and Sharpe, also underlined, are 14.30% and 0.566. These are descriptive comparisons conditional on historical selection. The price-index gap includes a dividend-convention difference; the CRSP and FF comparisons retain distributions on both sides. No alternative allocation exceeds Market-cap weight's CAGR.

Sample, QIS, and ridge GMVPs trail market-capitalization weighting. Bayes–Stein's annual mean-return difference is -1.74 percentage points, with Holm-adjusted $p = 0.266$. This result concerns the specified constrained hybrid, not expected-return optimization generally.

1) Interpretation of the Full Model Comparison: Table II summarizes the full-sample comparison of all 19 allocations. The following interpretation relates their objectives and selection exposures to that comparison, with the later regime results distinguishing cumulative growth, volatility, drawdown, and Sharpe. Success on one criterion does not establish success on the others. Unless a market index is explicitly named, the following comparisons are against market-capitalization weighting within the defensive selection. The characteristic exposures in Table VI and the return differences in Fig. 10 support economic interpretations, but do not identify causal mechanisms.

Market-capitalization and equal weighting require no return-moment estimates. Their competitive performance is consistent with much of the defensive exposure entering through selection. Market-capitalization weighting has the highest cumulative return among the 19 allocations in Bull I, Crisis I, and Shock I, whereas equal weighting gains 6.66 percentage points relative to it in Bull II. The latter spreads weight across the selected names without estimating covariance; its advantage in one expansion does not establish a general preference for equal weighting. Both share the low-risk selection that outperforms the broad market in Crisis I but trails it through the decline and rebound of Crisis II.

Constant-correlation GMVP and its Ledoit–Wolf variant restrict pairwise dependence and emphasize marginal volatility. This complements the low-volatility characteristic but can alter the balance with negative coskewness. For constant correlation, the low-volatility score rises by 2.04 percentile points while the coskewness score falls by 1.06 relative to the allocation benchmark. It gains 11.09 cumulative-return points in Recovery I, yet loses an additional 4.18 points in Crisis II. A covariance restriction estimated from trailing

TABLE II

FULL-SAMPLE PERFORMANCE USING OBSERVED CRSP TOTAL RETURNS WITH NO IMPUTATION AND A ROUND-TRIP TRADING COST OF 20 BASIS POINTS (10 BASIS POINTS PER DOLLAR PURCHASED OR SOLD). RETURNS, VOLATILITY, MDD, AND TURNOVER ARE PERCENTAGES. BOLD DENOTES FAVORABLE EXTREMA AMONG THE 19 STRATEGY ALLOCATIONS. UNDERLINING IDENTIFIES THE MARKET-CAPITALIZATION-WEIGHTED RETURN AND SHARPE BENCHMARKS AND THE LOWEST STRATEGY SHARPE RATIO. MARKET INDICES ARE EXCLUDED FROM THESE RANKINGS. THE BOLD S&P 500 LABEL AND UNDERLINED INDEX VALUES IDENTIFY THE PRIMARY PRICE BENCHMARK, NOT STATISTICAL SIGNIFICANCE.

Allocation	CAGR	Volatility	Sharpe	MDD	Turnover
LW identity	13.48	10.82	**1.114**	-31.16	19.49
Market-cap weight	**14.05**	11.39	1.110	**-29.00**	13.88
GMVP–1/N mixture	13.46	10.94	1.102	-31.55	17.00
Ridge GMVP	13.35	10.88	1.098	-31.88	20.14
Sample GMVP	13.36	10.91	1.096	-32.58	20.80
Norm GMVP	13.36	10.91	1.096	-32.58	20.80
Equal weight	13.52	11.18	1.087	-30.77	13.62
Inverse volatility	13.44	11.11	1.086	-30.80	13.29
QIS	13.18	10.87	1.085	-31.06	20.60
Inverse variance	13.37	11.08	1.083	-30.64	13.47
Transaction-cost GMVP	13.45	11.24	1.075	-33.28	**12.34**
Constrained risk budgeting	13.12	10.97	1.072	-31.04	14.98
Constant-corr. GMVP	13.20	11.08	1.069	-29.87	14.41
LW constant-corr.	13.18	11.08	1.068	-29.29	15.65
FF5 GMVP	12.80	10.86	1.055	-30.07	17.88
Minimum CVaR	12.97	11.13	1.046	-31.57	21.82
Maximum diversification	12.42	**10.71**	1.035	-30.62	23.95
RMT clipping	12.53	10.98	1.023	-31.60	19.01
Bayes–Stein Sharpe	12.15	10.97	0.995	-32.28	22.22
S&P 500 (price)	8.61	14.30	0.566	-52.56	–
CRSP market	10.86	14.93	0.687	-51.48	–
FF market	11.54	14.83	0.732	-50.31	–

observations need not capture abrupt changes in dependence or rebound participation. These outcomes are consistent with that limitation, not a test that estimation lag caused the loss. Neither covariance objective explicitly rewards the coskewness score, whose potential premium and adverse-state exposure must be distinguished from downside protection.

Sample GMVP, LW identity, ridge, and QIS retain more pairwise information than a pure constant-correlation target. In Recovery I their cumulative-return gains range from 12.82 to 15.85 points, and LW identity has the highest allocation Sharpe, 2.476. Identity shrinkage, ridge, and QIS regularize weak sample directions differently; these gains do not identify which estimator forecasts covariance most accurately. RMT clipping and FF5 GMVP impose spectral and factor restrictions, respectively, and gain less in that recovery. Their stronger structural assumptions trade estimation variance against loss of relevant dependence. FF5 nevertheless has the lowest realized volatility in Crisis I, 15.44%, while its cumulative loss exceeds the allocation benchmark's: successful risk reduction need not maximize terminal wealth. Norm-constrained GMVP coincides exactly with sample GMVP because its norm bound is redundant, so it supplies no distinct regime mechanism.

Inverse volatility, inverse variance, constrained risk budgeting, and the GMVP–$1/N$ mixture all outperform market-capitalization weighting in Bull II. The first two emphasize marginal-risk differences, risk budgeting balances estimated risk contributions, and mixing tempers covariance-driven concentration. Their broader holdings provide alternatives to concentrated minimum variance within an already low-risk universe; they do not directly optimize coskewness compensation. All four trail the allocation benchmark in Shock I and Recovery II, so diversification alone does not explain a stable return advantage.

Maximum diversification targets the ratio of weighted marginal volatility to portfolio volatility, not the lowest individual-stock volatility. It has the highest allocation Sharpe in Bull II and is the only alternative with a positive cumulative-return difference in Recovery II, but its high turnover and gross-return shortfall weaken full-sample performance. Minimum CVaR instead targets the empirical loss tail. In Recovery II it has the highest allocation Sharpe (3.082) and least severe MDD (-1.81%), despite lower cumulative growth than the allocation benchmark. Its additional losses in both crisis windows and effective tail sample of only 3.6 observations limit any claim of reliable crisis insurance. Bayes–Stein Sharpe introduces estimated expected returns, with shrinkage intended to temper their uncertainty. It trails the allocation benchmark in cumulative return in all nine regimes; higher fitted means therefore fail to translate into incremental realized growth. None of these objectives explicitly preserves the original three-characteristic exposure balance.

Transaction-cost GMVP regularizes departures from pretrade holdings. It leads cumulative growth in Bull III by 7.07 points relative to the allocation benchmark, but has a slightly lower Sharpe. Retaining existing exposures can affect gross performance as well as trading costs; turnover-matched comparisons are needed to separate those channels. The full-sample cost saving remains smaller than the gross-return sacrifice. Its economic value consequently depends on the cost level and the return consequences of slower reallocation, not solely on lower turnover.

Figure 8 aligns log wealth and drawdown for three representative allocations and all three market benchmarks. The thick orange line emphasizes S&P 500; its label states the price-return convention. Equal proportional wealth changes occupy equal vertical distances, and monthly drawdowns omit intramonth extremes.

2) Paired Statistical Evidence: All 18 annualized mean net-return differences from market-capitalization weighting are negative and insignificant after Holm adjustment (Table III, Panel A). Constant-correlation GMVP trails by 0.79 percentage points, with HAC statistic -0.83, adjusted $p = 1.000$, and Sharpe-difference interval $[-0.232, 0.143]$. The underlined entries highlight the absence of evidence for an incremental advantage.

LW identity's Sharpe advantage is only 0.004, with paired 95% interval $[-0.182, 0.160]$. Its lower CAGR and interval spanning zero do not support selecting it from the observed ranking.

3) Market-Relative Inference and Temporal Stability: Panel B of Table III addresses whether the selected portfolio outperforms the market, a different question from whether reweighting improves on that portfolio. Over February 2003–December 2020, market-capitalization weighting has Sharpe 1.110 versus 0.732 for the FF market. The bold $\Delta SR = 0.379$ and 95% interval $[0.079, 0.720]$ support a positive gap at the nominal 5% level when the retained specification is treated as fixed. The bold annualized FF5 alpha is 5.67%, with HAC statistic 3.99 and two-sided $p < 0.001$. This evidence is stronger than a comparison of point estimates alone, but remains conditional on the return model and historical strategy choice.

The full-sample Sharpe differences are also positive against CRSP (0.423; interval $[0.128, 0.762]$) and S&P 500 (0.544; $[0.250, 0.885]$). The S&P comparison retains the price-versus-total-return mismatch; its larger gap is not evidence of a larger dividend-matched premium. The FF comparison preserves the original study's market definition. In that comparison, 12 of 2,000 unrestricted bootstrap differences are nonpositive (0.6%). This tail frequency is not a null-imposed one-sided p value or a probability that the strategy's population advantage is absent.

The temporal decomposition qualifies the full-sample result. Market-cap weight's Sharpe changes little in point estimates, from 1.093 in 2003–2011 to 1.123 in 2012–2020, while the FF market's rises from 0.417 to 1.085. The gap therefore declines from 0.676 to 0.038. The underlined later-period interval $[-0.351, 0.495]$ includes zero; FF5 alpha falls to 2.25%, with HAC statistic 1.22 and underlined $p = 0.222$. Failure to reject is not evidence that the true advantage is exactly zero: the interval admits economically meaningful gains and losses.

Three- and twelve-month bootstrap blocks preserve the positive full-sample interval and the zero-containing 2012–2020 interval. The point estimates are consistent with an advantage concentrated in the earlier, crisis-affected period and a stronger market benchmark thereafter. They neither identify the market regime as a causal explanation nor constitute a formal test of a structural break. Stable strategy Sharpe across two retrospectively defined periods does not exclude overfitting, and significance in one period but not another does not itself establish a statistically significant change. An untouched evaluation sample remains necessary to assess persistence after specification selection.

B. Market-Regime Subperiods

The nine regimes in Table IV partition all 215 observed months. February 2020 belongs only to Crisis II; the bold Pre-Crisis boundary ends in January. Bull I starts in February 2003 and Recovery II ends in December 2020. Monthly counts are 53, 21, 39, 36, 9, 33, 13, 5, and 6. Underlined counts flag the imprecision of annualized estimates from the two shortest periods.

Regimes serve only ex-post evaluation and do not enter portfolio decisions. We calculate monthly sample counts and returns, omitting unverifiable ancillary statistics from the reference calendar. Crisis II covers February–June 2020 and therefore includes the April–June rebound.

Table V compares market-capitalization weighting, equal weighting, constant-correlation GMVP, and all three market benchmarks. S&P 500 labels are bold and its values underlined in every regime; these identify the primary reference, not favorable rankings. Other bold values identify the best cumulative return and Sharpe among the three displayed strategies. Supplementary results cover all 19 rules under six cost scenarios. Regime statistics use continuous returns without retuning or boundary costs. MDD resets wealth at each regime start; annualized short-period measures should be read alongside cumulative returns and sample counts.

The underlined S&P regime returns show why the full-sample ranking is insufficient. All 19 allocations exceed its -46.93% return in Crisis I, but none exceeds its -3.88% in Crisis II or its 51.46% gain in Bull II. Only five exceed S&P in Pre-Crisis and six in Recovery II. These counts refer to the price benchmark and the specified windows; they do not establish dividend-matched alpha or general crisis protection.

During Crisis I, market-capitalization weighting returns -20.77% versus CRSP's -44.85%. Constant correlation has lower volatility (15.92% versus 18.42%), MDD -27.03%, and the underlined return of -23.93%. Return, volatility, and drawdown must be read jointly: with negative average excess returns, lower volatility alone can make Sharpe more negative.

Relative rankings vary. Constant correlation exceeds market-capitalization weighting in Recovery I (96.36% versus 85.27%) but trails in Bull II (38.94% versus 41.17%; CRSP 56.59%). In Shock I, market-capitalization weighting and constant correlation gain 10.49% and 9.41%, versus CRSP -2.26%. Other bold maxima are Market-cap weight's Bull I return (141.16%) and equal weighting's Sharpe (2.445); constant correlation's Bull III return (38.17%) and Market-cap weight's Sharpe (1.207); and constant correlation's Pre-Crisis return (30.91%) and Sharpe (2.672).

In Crisis II, market-capitalization weighting and constant correlation return -11.73% and -15.91%, versus CRSP -3.41% (Sharpe -0.087). The underlined constant-correlation loss contrasts with the earlier crisis. During July–December 2020, their returns are 21.57%, 19.19%, and 25.22%, with Sharpes 2.296, 2.721, and 2.323. Bold entries distinguish cumulative-growth and Sharpe leaders. Crisis II spans both the

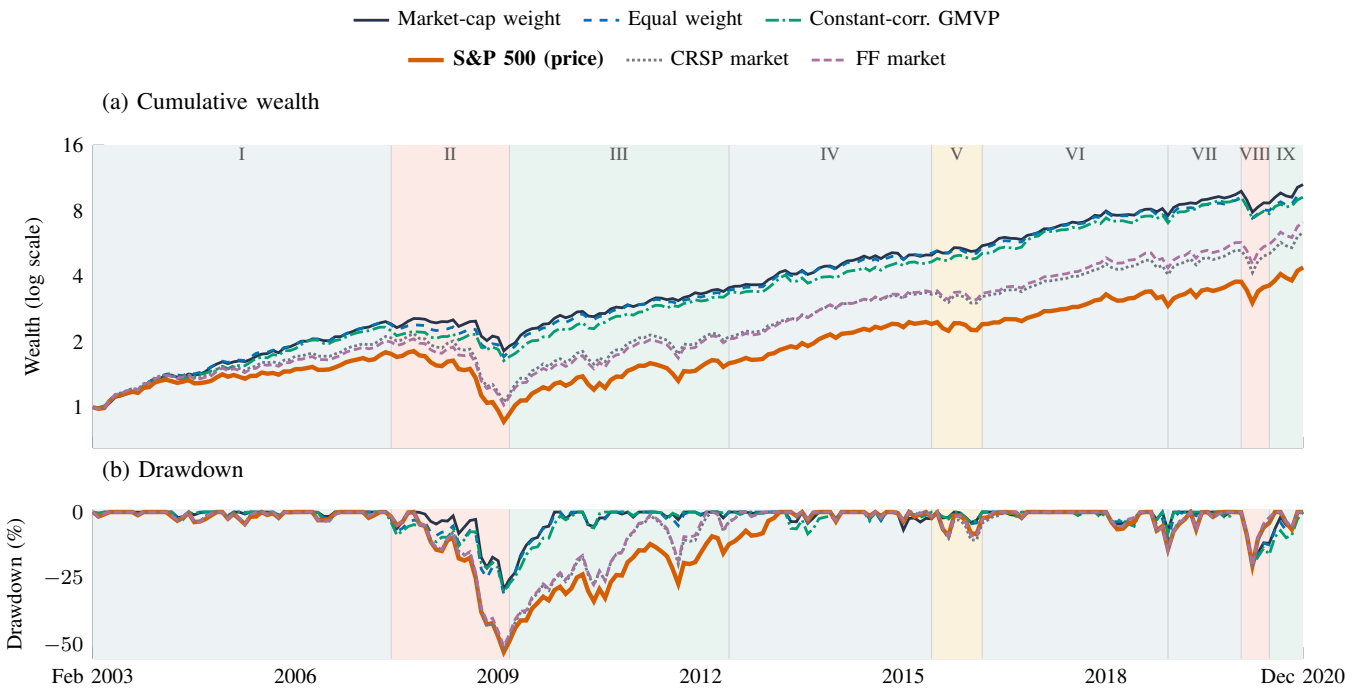


Fig. 8. Wealth and drawdown, February 2003–December 2020. Log wealth starts at one; drawdown includes the initial peak. Bands mark nine regimes, with crises in pale red. The thick orange line highlights the primary S&P 500 price benchmark, which excludes dividends; CRSP and FF include distributions. Strategies include 20-basis-point round-trip costs; indices are uncosted. Constant correlation illustrates a structural covariance rule independently of its performance ranking.

TABLE III

ALLOCATION AND MARKET-RELATIVE INFERENCE. PANEL A COMPARES ALLOCATIONS WITH MARKET-CAPITALIZATION WEIGHTING; MEAN DIFFERENCES ARE ANNUALIZED PERCENTAGE POINTS AND NO MEAN DIFFERENCE IS SIGNIFICANT AFTER HOLM ADJUSTMENT. UNDERLINING HIGHLIGHTS CONSTANT CORRELATION. PANEL B COMPARES MARKET-CAPITALIZATION WEIGHTING (P) WITH THE FF TOTAL-RETURN MARKET (M); n COUNTS MONTHS AND ALPHA IS THE ANNUALIZED FF5 INTERCEPT. SHARPE INTERVALS USE 2,000 PAIRED SIX-MONTH BLOCK-BOOTSTRAP DRAWS; ALPHA STATISTICS USE HAC(4). BOLD IDENTIFIES FULL-SAMPLE EVIDENCE AND UNDERLINING THE LATER-PERIOD INTERVAL AND ALPHA p VALUE. MARKET INFERENCE IS POINTWISE AND UNADJUSTED FOR HISTORICAL SPECIFICATION SEARCH.

Panel A. Incremental allocation performance

Allocation	Mean difference	HAC statistic	Holm p	ΔSharpe	Sharpe interval
Equal weight	-0.49	-0.72	1.000	-0.024	[-0.159, 0.111]
Constant-corr. GMVP	-0.79	-0.83	1.000	-0.041	[-0.232, 0.143]
Transaction-cost GMVP	-0.55	-0.62	1.000	-0.036	[-0.220, 0.130]
Inverse volatility	-0.57	-0.87	1.000	-0.025	[-0.153, 0.112]
Sample GMVP	-0.67	-0.74	1.000	-0.014	[-0.220, 0.148]
Bayes–Stein Sharpe	-1.74	-2.44	0.266	-0.116	[-0.249, 0.025]

Panel B. Market-relative performance and FF5 alpha

Period	n	SR_P	SR_M	ΔSR	95% CI	$12\widehat{\alpha}$ (%)	t_{HAC}	p_{HAC}
2003–2020	215	1.110	0.732	**+0.379**	**[+0.079, +0.720]**	**5.67**	**3.99**	< 0.001
2003–2011	107	1.093	0.417	+0.676	[+0.300, +1.255]	9.69	5.04	< 0.001
2012–2020	108	1.123	1.085	+0.038	[-0.351, +0.495]	2.25	1.22	0.222

abrupt pandemic decline and the subsequent V-shaped market rebound; its five-month return is not a pure downside-protection test. Underperformance over that window limits any claim of protection across crises, even though the strategy's full-sample volatility is lower.

These descriptive comparisons do not independently test allocation skill or determine rule selection. Five- and six-month intervals provide particularly imprecise annualized estimates.

Figure 9 rebases each regime to unit wealth with common vertical scales and portfolio styles. Terminal values show cumulative growth without annualizing short samples; equal panel widths do not imply equal durations. These paths complement the full-history drawdown in Figure 8.

Figure 10 includes every alternative allocation and all three market series, with the S&P label in bold. For allocation or index a in regime g, its difference from market-capitalization weighting is

$$\Delta_{a,g} = 100 \left\{ \prod_{t \in g} (1 + R_t^a) - \prod_{t \in g} (1 + R_t^0) \right\}, \qquad (40)$$

TABLE IV
MARKET REGIMES WITHIN THE EVALUATION SAMPLE, FEBRUARY 2003–DECEMBER 2020. THE SAMPLE SIZE n COUNTS MONTHLY HOLDING-PERIOD RETURNS. BOLD IDENTIFIES THE BOUNDARY BETWEEN PRE-CRISIS AND CRISIS II; UNDERLINING IDENTIFIES THE TWO SHORTEST SAMPLES.

Regime	Type	Evaluation period	n
Bull I	Bull	2003-02–2007-06	53
Crisis I	Crisis	2007-07–2009-03	21
Recovery I	Recovery	2009-04–2012-06	39
Bull II	Bull	2012-07–2015-06	36
Shock I	Shock	2015-07–2016-03	9
Bull III	Bull	2016-04–2018-12	33
Pre-Crisis	Bull	**2019-01–2020-01**	13
Crisis II	Crisis	**2020-02–2020-06**	<u>5</u>
Recovery II	Recovery	2020-07–2020-12	<u>6</u>

where 0 denotes market-capitalization weighting. Cells measure cumulative-return differences in percentage points, not annualized returns or relative wealth growth. A common zero-centered scale groups rows by methodology; market rows are separated because they exclude strategy costs.

All 19 allocations outperform CRSP in Crisis I: cumulative returns range from -28.59% to -20.77%, versus -44.85%. None outperforms in Crisis II: returns range from -15.91% to -9.81%, versus -3.41%. These window-dependent outcomes, including the 2020 rebound, preclude a general claim of crisis protection.

Constant correlation exceeds market-capitalization weighting only in Recovery I, Bull III, and Pre-Crisis, by 11.09, 0.55, and 1.68 percentage points. In Recovery II, 17 of 18 alternatives trail the benchmark. Heatmap differences are descriptive, not significance tests, and unequal regime lengths preclude direct comparison as return rates.

C. Additional Results: Mechanisms and Robustness

1) Covariance Structure of Selected Stocks: Common variation and marginal-risk heterogeneity remain after selection. Average pairwise correlation is 0.303, with cross-pair dispersion 0.176. Median leading-eigenvalue variance share is 33.7%, covariance condition number approximately 3,581, and the 90th/10th-percentile volatility ratio 1.66. Estimating 435 correlations from 36 months invites error, but their heterogeneity also makes structural restrictions potentially costly.

Within selected stocks, score-rank correlations average 0.03 for beta–volatility and -0.29 for both coskewness pairs. These differ from full-universe estimates: conditioning on a high composite score can induce negative component associations as one offsets another. Selected-sample correlations cannot establish full-universe characteristic independence.

2) Portfolio-Weighted Characteristic Exposures: Constant correlation raises the portfolio-weighted low-volatility percentile by 2.04 percentage points and lowers the coskewness percentile by 1.06, shown in bold and underline in Table VI. Fitted beta falls from market-capitalization weighting's 0.60 to 0.50. Common constituents therefore do not preserve the characteristic balance established by selection.

Effective holdings are 16.9 for constant correlation, 19.8 for market-capitalization weighting, and 30 for equal weighting. Constant correlation averages 10.2 capped positions; sample GMVP has 16.3 effective holdings and 11.6 caps. Fitted volatilities for sample, constant correlation, and Market-cap weight are 8.21%, 9.19%, and 10.35%, versus realized 10.91%, 11.08%, and 11.39%. Conditional fitted and unconditional realized risk are distinct, so these gaps are not calibration tests.

3) Gross Performance and Trading Costs: For each allocation rule, the difference in annualized mean net returns relative to the market-capitalization-weighted benchmark satisfies

$$12\overline{(R^a_{net} - R^0_{net})} = 12\overline{(R^a_{gross} - R^0_{gross})} - 12\overline{(C^a - C^0)}, \tag{41}$$

where $C = R_{gross} - R_{net}$ and 0 denotes market-capitalization weighting. This is an accounting identity, not a causal decomposition. Constant correlation's annual net difference of -0.79 percentage points comprises -0.77 before costs and 0.01 additional costs. LW constant correlation has differences of -0.76 gross and -0.81 net. Gross returns dominate both shortfalls.

Sample GMVP trails by 0.67 annual percentage points: 0.50 gross and 0.17 additional costs, approximately 25% of the shortfall. Bold entries in Table VI mark its cost and net differences; positive ΔCost means additional cost. LW identity, ridge, and QIS add 0.14, 0.15, and 0.16 points, respectively. Covariance regularization need not reduce trading.

RMT's gross shortfall is 1.28 percentage points, with 0.12 additional costs. FF5's fitted beta of 0.42 and volatility of 8.66% coexist with a 1.08-point gross shortfall and 0.10 additional costs. Lower estimated exposure thus does not ensure higher realized returns.

4) Risk-Based Allocation and Transaction-Cost Effects: Inverse volatility and inverse variance average 28.7 and 26.1 effective holdings, with annual net shortfalls of 0.57 and 0.64 percentage points. Their relatively low turnover and broad holdings do not yield higher CAGR or Sharpe than market-capitalization weighting.

Risk budgeting averages 27.4 effective holdings and 0.6 active caps. Risk-contribution dispersion is 0.0008 versus equal weighting's 0.0108; binding-cap deviations follow Section III-E2. The GMVP–$1/N$ mixture raises effective holdings to 24.8 but trails market-capitalization weighting gross by 0.50 annual percentage points. The norm rule remains identical to sample GMVP.

Transaction-cost GMVP saves 0.04 annual percentage points but sacrifices 0.59 gross, leaving a 0.55 net shortfall. Bold cost and net entries mark this trade-off; negative ΔCost denotes savings. Turnover falls to 12.34% from Market-cap weight's 13.88% and sample GMVP's 20.80%, but savings at 20 basis points round trip do not offset lower gross returns.

5) Expected-Return Estimation and Tail-Risk Optimization: CVaR averages 15.6 effective holdings and 13.2 caps. Its annual net shortfall of 0.99 percentage points comprises 0.80 gross and 0.19 additional costs. The 3.6-observation empirical tail limits estimation precision despite accurate optimization.

Maximum diversification sacrifices 1.29 annual percentage points gross and adds 0.24 in costs. Bayes–Stein fits a monthly

TABLE V

REGIME PERFORMANCE AT BASELINE COSTS, ORDERED CHRONOLOGICALLY DOWN THE LEFT PANEL AND THEN THE RIGHT. DATES AND MONTHLY COUNTS APPEAR IN TABLE IV. CUMULATIVE RETURN, CAGR, VOLATILITY, AND MDD ARE PERCENTAGES; CC DENOTES CONSTANT-CORRELATION GMVP. BOLD MARKS THE HIGHEST STRATEGY CUMULATIVE RETURN AND SHARPE, AND IDENTIFIES THE PRIMARY S&P 500 BENCHMARK. UNDERLINING MARKS S&P VALUES AND CC CRISIS RETURNS. INDICES ARE UNCOSTED; S&P EXCLUDES DIVIDENDS, WHEREAS CRSP AND FF INCLUDE THEM.

Portfolio	Cum. ret.	CAGR	Vol.	Sharpe	MDD
Bull I					
Market-cap weight	**141.16**	22.06	7.27	2.393	-2.49
Equal weight	139.18	21.83	7.01	**2.445**	-2.96
CC GMVP	124.17	20.05	7.36	2.137	-3.89
S&P 500 (price)	75.69	13.61	8.21	1.257	-4.54
CRSP market	108.18	18.06	8.84	1.610	-4.60
FF market	98.95	16.85	8.72	1.512	-4.65
Crisis I					
Market-cap weight	**-20.77**	-12.46	18.42	**-0.743**	-29.00
Equal weight	-24.46	-14.81	17.78	-0.927	-28.66
CC GMVP	-23.93	-14.47	15.92	-1.032	-27.03
S&P 500 (price)	-46.93	-30.37	20.88	-1.714	-52.56
CRSP market	-44.85	-28.83	22.04	-1.513	-51.48
FF market	-43.87	-28.10	21.23	-1.534	-50.31
Recovery I					
Market-cap weight	85.27	20.89	9.25	2.105	-5.44
Equal weight	90.92	22.02	9.24	2.210	-5.67
CC GMVP	**96.36**	23.07	8.70	**2.442**	-5.98
S&P 500 (price)	70.72	17.89	16.30	1.092	-17.03
CRSP market	83.48	20.53	17.37	1.164	-18.50
FF market	84.08	20.65	16.87	1.199	-17.65
Bull II					
Market-cap weight	41.17	12.18	9.57	1.251	-6.71
Equal weight	**47.82**	13.92	9.05	**1.489**	-4.37
CC GMVP	38.94	11.59	9.79	1.171	-8.22
S&P 500 (price)	51.46	14.84	8.50	1.677	-3.56
CRSP market	56.59	16.12	8.51	1.806	-3.07
FF market	64.21	17.98	8.92	1.907	-3.30
Shock I					
Market-cap weight	**10.49**	14.23	11.02	**1.255**	-3.93
Equal weight	6.21	8.36	10.63	0.797	-4.45
CC GMVP	9.41	12.75	10.25	1.213	-4.03
S&P 500 (price)	-0.16	-0.22	16.94	0.057	-8.74
CRSP market	-2.26	-3.00	16.90	-0.111	-9.85
FF market	-1.10	-1.46	17.08	-0.016	-8.97

Portfolio	Cum. ret.	CAGR	Vol.	Sharpe	MDD
Bull III					
Market-cap weight	37.62	12.31	9.12	**1.207**	-7.08
Equal weight	33.82	11.18	9.68	1.036	-10.04
CC GMVP	**38.17**	12.48	10.68	1.058	-7.99
S&P 500 (price)	21.71	7.40	10.21	0.651	-13.97
CRSP market	26.07	8.79	10.17	0.778	-14.17
FF market	30.86	10.27	10.66	0.875	-14.53
Pre-Crisis					
Market-cap weight	29.24	26.71	8.84	2.512	-1.51
Equal weight	26.03	23.81	9.78	2.037	-1.43
CC GMVP	**30.91**	28.23	8.79	**2.672**	-1.88
S&P 500 (price)	28.67	26.20	12.61	1.756	-6.58
CRSP market	29.09	26.58	12.88	1.747	-6.17
FF market	30.59	27.94	13.48	1.754	-6.71
Crisis II					
Market-cap weight	**-11.73**	-25.88	27.52	**-0.976**	-19.87
Equal weight	-12.35	-27.11	25.83	-1.116	-19.12
CC GMVP	-15.91	-34.03	25.77	-1.492	-20.12
S&P 500 (price)	-3.88	-9.07	35.13	-0.145	-19.87
CRSP market	-3.41	-7.99	37.34	-0.087	-20.86
FF market	-1.94	-4.59	37.20	0.007	-20.20
Recovery II					
Market-cap weight	**21.57**	47.80	17.80	2.296	-4.38
Equal weight	21.20	46.90	14.96	2.665	-3.28
CC GMVP	19.19	42.05	13.31	**2.721**	-2.88
S&P 500 (price)	21.15	46.78	19.80	2.044	-6.58
CRSP market	25.22	56.79	20.40	2.323	-5.45
FF market	26.50	60.03	20.98	2.365	-5.62

mean of 1.36%, versus Market-cap weight's 1.20%, yet trails gross by 1.54 points and adds 0.20 in costs. Numerical optimality does not establish forecast quality or allocation stability.

Independent checks cover covariance formulas, convex optimality, and ratio and risk budgeting solutions (Appendix A).

6) Trading Costs: Table VII holds baseline target weights fixed as costs vary. Market-cap weight's CAGR declines from 14.43% before costs to 14.05% at 10 basis points per dollar traded and 10.67% at 100. The underlined 10-basis-point row represents 20 basis points round trip; bold marks the best displayed CAGR at each rate. These are accounting sensitivities, without recalibrating the trading penalty.

Proportional costs do not estimate bid–ask spreads, market impact, or capacity. The size filter may improve feasibility, but its separate effect requires excluded-stock and complete capitalization data.

7) Source-Return Recovery and Historical Reconciliation: Earlier apparent market underperformance arose from omitted source returns and changed cost assumptions. The filtered extract excluded 1,109 available returns; the workflow substituted 1,086 price changes and imposed 23 terminal losses. Restoring observed returns invalidates that interpretation. All current allocations are estimated and evaluated on the restored data.

Table VIII reconciles annualization conventions. At 20 basis points round trip, Market-cap weight has CAGR 14.05%, while $(1 + \overline{R})^{12} - 1 = 14.79\%$ matches the original 14.8% after rounding. Sharpe 1.110 and MDD -29.00% similarly match 1.11 and -29.0%. This agreement does not independently reproduce selection or every original accounting entry.

With original Market-cap weight targets held fixed, restoring returns gives CAGR 12.54% at the previous cost of 50 basis points per dollar traded; reducing that rate to 10 gives 14.05%. Neither change reflects optimization. Other rules

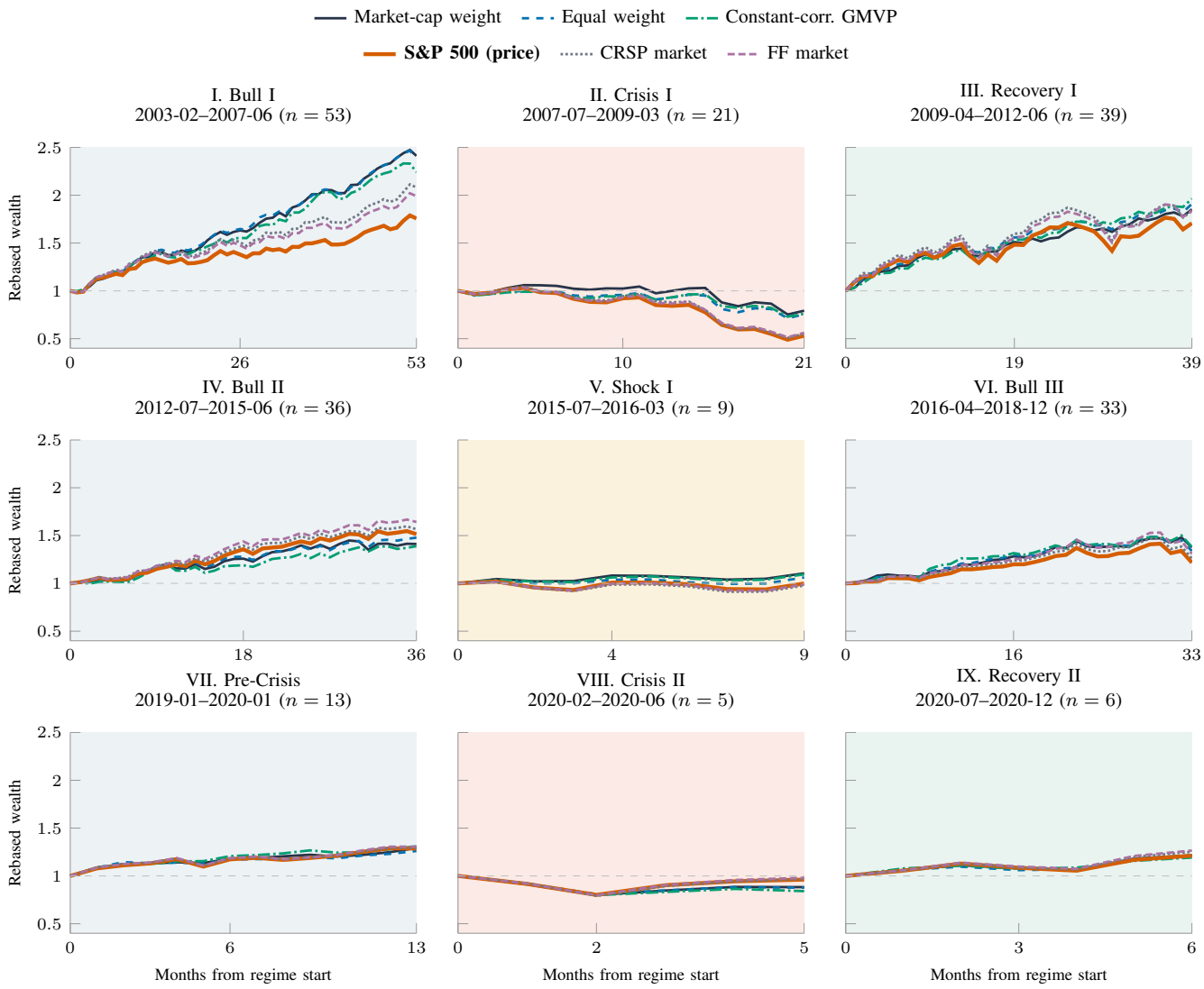


Fig. 9. Regime wealth rebased to one, with common vertical scales and elapsed-month axes. The thick orange S&P 500 line and other styles match Fig. 8. Strategies include 20-basis-point round-trip costs; all indices are uncosted, and only S&P 500 excludes dividends. Bull I starts in February 2003 and Recovery II ends in December 2020. February 2020 belongs only to Crisis II. Unequal durations and short samples limit comparisons.

are reestimated from restored windows, with transaction-cost GMVP recalibrated to $\lambda = 0.0004$. Historical results and the reconciliation are archived separately.

8) Return-Oriented versus Defensive Selection: We separately compare the supplied momentum–profitability–issuance selection (50 stocks) with the low-risk–coskewness defensive selection (30 stocks). Both use February 2003–December 2020, observed CRSP total returns, drift-adjusted transaction costs of 10 basis points per dollar traded, and terminal proceeds held in cash. Constituents are retained exactly; no new size filter or missing-return imputation is applied. The two allocation benchmarks are equal weighting and recorded market-capitalization weights, with the common rule $u = 2/N$ giving limits of 4.00% and 6.67%, respectively. Reconstructed proportional weights agree with both supplied files within 5×10^{-7}, and the defensive monthly net returns reproduce the article baseline within 10^{-12}.

Under market-capitalization weighting, CAGR rises by 2.43 percentage points, but volatility rises by 7.06 points and Sharpe falls from 1.110 to 0.858. Maximum drawdown becomes 19.04 percentage points more severe. Equal weighting gives the same qualitative result. Thus the observed growth gain comes with materially greater realized risk; the new selection does not establish a return improvement at comparable risk.

For market-capitalization weights, a paired six-month circular block bootstrap (2,000 draws) gives pointwise 95% intervals of $[-4.46, 9.58]$ percentage points for the CAGR difference, $[5.18, 8.91]$ points for the volatility difference, and $[-0.628, 0.073]$ for the Sharpe difference. All differences are return-oriented minus defensive. The volatility increase is clear, whereas the CAGR interval includes zero. The paired mean-return HAC(4) test has $p = 0.312$, or $p = 0.401$ after Holm adjustment across the two weighting rules. A Sharpe interval containing zero is not an equivalence test or evidence that risk is unchanged.

In 2012–2020, equal-weight CAGR is 12.85% for the return-

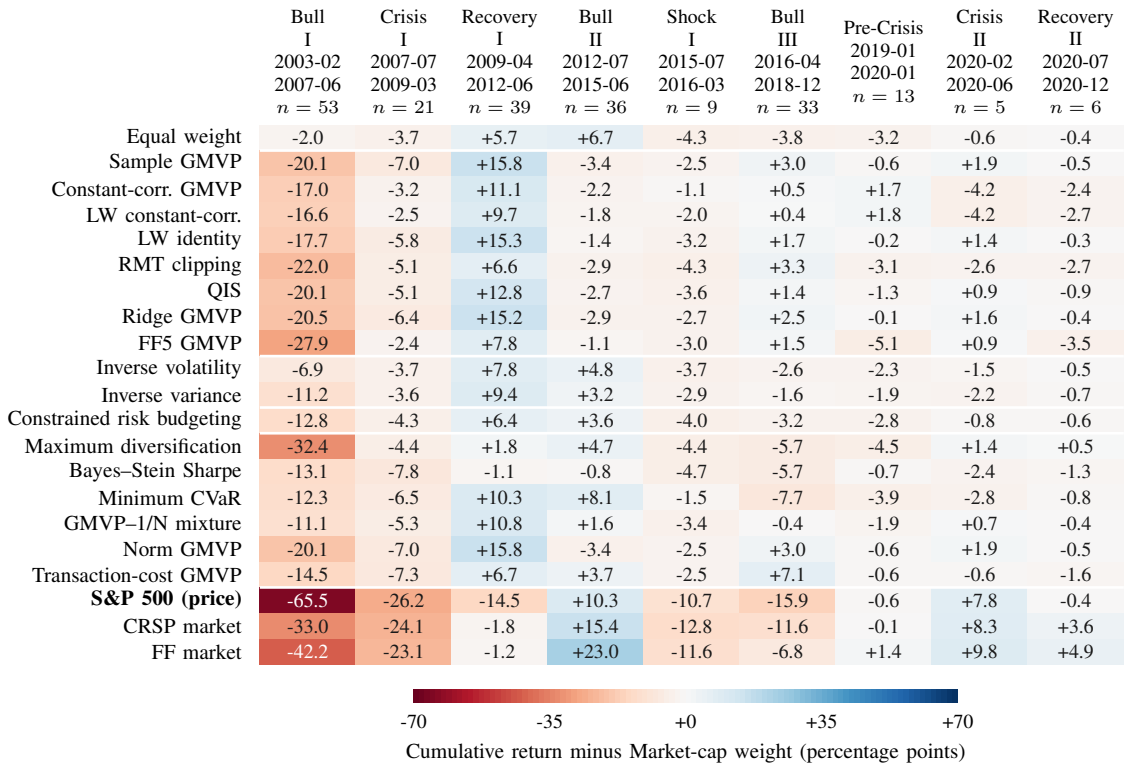


Fig. 10. Cumulative-return differences from market-capitalization weighting, in percentage points. Blue denotes higher terminal wealth and red lower wealth on a common symmetric scale. Rows follow methodological groups; the bold S&P 500 label highlights the primary price benchmark, not significance. Headings give observed dates and monthly counts. Strategies include 20-basis-point round-trip costs; market indices are uncosted. S&P 500 excludes dividends, whereas CRSP and FF include them. Unequal regime lengths limit comparisons across columns; colors do not indicate significance.

TABLE VI

CONCENTRATION, CHARACTERISTIC EXPOSURES, AND RETURN DECOMPOSITION. HOLDINGS AND CAP COUNTS ARE MONTHLY AVERAGES; B/V/C ARE PORTFOLIO-WEIGHTED PERCENTILE DIFFERENCES FROM MARKET-CAPITALIZATION WEIGHTING. RETURN AND COST DIFFERENCES ARE ANNUALIZED ARITHMETIC PERCENTAGE POINTS, NOT CAGR DIFFERENCES. BOLD HIGHLIGHTS DISCUSSED VOLATILITY AND COST EFFECTS; UNDERLINING MARKS LOWER COSKEWNESS EXPOSURE.

Allocation	N_{eff}	At cap	B tilt	V tilt	C tilt	ΔGross	ΔCost	ΔNet
Market-cap weight	19.8	7.5	0.00	0.00	0.00	0.00	0.00	0.00
Equal weight	30.0	0.0	-0.27	-0.49	0.56	-0.50	-0.01	-0.49
Sample GMVP	16.3	11.6	0.51	1.05	-1.32	-0.50	**0.17**	**-0.67**
Constant-corr. GMVP	16.9	10.2	0.38	**2.04**	-1.06	-0.77	0.01	-0.79
LW constant-corr.	17.0	10.0	0.40	1.98	-1.11	-0.76	0.04	-0.81
LW identity	18.2	7.7	0.52	0.99	-1.22	-0.43	0.14	-0.57
RMT clipping	16.9	10.4	0.59	1.43	-1.24	-1.28	0.12	-1.40
QIS	17.4	9.0	0.58	1.01	-1.25	-0.67	0.16	-0.83
Ridge GMVP	17.1	9.5	0.51	1.04	-1.28	-0.53	0.15	-0.68
FF5 GMVP	17.4	9.1	0.84	1.15	-1.07	-1.08	0.10	-1.17
Inverse volatility	28.7	0.0	-0.13	0.13	0.13	-0.58	-0.01	-0.57
Constrained risk budgeting	27.4	0.6	0.02	0.07	-0.12	-0.84	0.03	-0.87
Maximum diversification	16.4	11.3	0.33	-0.24	-0.93	-1.29	0.24	-1.53
Bayes–Stein Sharpe	17.6	9.0	0.32	0.64	-0.81	-1.54	0.20	-1.74
Minimum CVaR	15.6	13.2	0.50	0.86	-1.37	-0.80	0.19	-0.99
GMVP–1/N mixture	24.8	0.0	0.12	0.28	-0.38	-0.50	0.08	-0.58
Norm GMVP	16.3	11.6	0.51	1.05	-1.32	-0.50	0.17	-0.67
Transaction-cost GMVP	16.8	7.7	0.64	0.70	-0.83	-0.59	**-0.04**	**-0.55**
Inverse variance	26.1	0.7	0.01	0.67	-0.24	-0.65	-0.01	-0.64

oriented selection versus 12.98% for the defensive selection. Market-capitalization weighting retains a CAGR increase (14.99% versus 13.56%), but volatility remains higher (15.86% versus 11.40%) and Sharpe lower (0.924 versus 1.123). The comparison is conditional on historical selections: different stock counts, associated position limits, and prior factor searches prevent attributing the difference solely to the factors. Optimization may still improve risk-adjusted performance within the new selection, but that is a separate comparison against its own equal-weight baseline.

9) Interpretation and Remaining Limits: Selection plausibly accounts for much of the risk reduction: substantial defensive performance is present before replacing market-capitalization weights with an optimizer. This is a conditional interpretation, not a causal decomposition. Common monthly constituent sets isolate changes in allocation but prevent identifying separate

TABLE VII
CAGR (%) UNDER ALTERNATIVE PROPORTIONAL TRANSACTION COSTS, HOLDING PORTFOLIO WEIGHTS AND OBSERVED TOTAL RETURNS CONSTANT. MC DENOTES MARKET-CAPITALIZATION WEIGHTING WITH THE COMMON POSITION LIMIT, CC CONSTANT-CORRELATION GMVP, AND TC TRANSACTION-COST GMVP. BOLD DENOTES THE HIGHEST CAGR AMONG THE FOUR DISPLAYED PORTFOLIOS AT EACH COST; UNDERLINING DENOTES THE BASELINE COST.

One-way bp	MC	Equal	CC	TC
0	**14.43**	13.90	13.60	13.78
<u>10</u>	**14.05**	13.52	13.20	13.45
25	**13.48**	12.97	12.62	12.94
50	**12.54**	12.05	11.65	12.11
75	**11.60**	11.13	10.68	11.28
100	**10.67**	10.22	9.73	10.45

TABLE VIII
RETURN-SOURCE RECONCILIATION AND CORRECTED MARKET-CAPITALIZATION-WEIGHTED PERFORMANCE. ALL PERFORMANCE MEASURES USE FEBRUARY 2003–DECEMBER 2020. BOLD IDENTIFIES THE CORRECTED RETURN MEASURES; NO ASSUMED LOSS IS USED.

Quantity	Value
Observed holding returns	6,450/6,450
Distinct observed estimation inputs	28,470/28,470
Restored holding returns	1,109
Restored termination observations	23
Round-trip cost (bp)	20
CAGR (%)	**14.05**
$(1+\overline{R})^{12}-1$ (%)	**14.79**
Sharpe ratio	1.110
MDD (%)	-29.00

beta, volatility, and coskewness effects. That attribution would require portfolios formed after removing or varying each selection characteristic under the same sample and trading assumptions.

Both three- and five-factor alphas are model-dependent and may reflect omitted risks or historical selection. In particular, FF5 does not explicitly include BAB or a low-volatility factor [5], [9]. The positive full-sample intercept could therefore represent implementation of a known low-risk anomaly rather than a distinct coskewness premium or managerial skill. Augmented factor regressions and counterfactual selection portfolios would be needed to distinguish these explanations. The weaker post-2011 estimates further limit claims of a persistent market-relative advantage.

Estimating 30-stock covariance from 36 months remains uncertain despite position constraints: there are 30 variances and 435 distinct pairwise covariances, or 465 unique entries, estimated from the same short history. Demeaning leaves 35 temporal degrees of freedom. The matrix can be full rank, but invertibility does not imply precise covariance estimates. Verified return coverage resolves the extraction error, while covariance restrictions still yield no higher CAGR than Market-cap weight. The maximum incremental Sharpe gain of 0.004 has a confidence interval spanning zero. These findings support limited evidence of allocation benefits in this selected universe; they neither establish that optimization is generally ineffective nor identify estimation error as the cause.

Three design limitations remain. First, missing upstream inputs prevent independent reconstruction of the selection stage, although allocation and return accounting can be reproduced conditional on the supplied constituent history. Second, constant proportional costs omit spreads, market impact, and partial-month cash remuneration; cost sensitivity at fixed weights does not establish implementable capacity. Third, historical searches across factors, portfolio size, and other specifications preclude an untouched signal-discovery test. Return recovery does not resolve these limitations.

The reported market confidence intervals and alpha p values are nominal, treat the retained strategy as fixed, and do not incorporate that search. Holm adjustment protects the stated family of 18 allocation mean-return comparisons; it does not retrospectively correct the broader strategy-development process. Consequently, the full-sample statistical evidence supports a historical association under the stated assumptions, not a search-adjusted discovery claim. A prespecified strategy evaluated on previously unused data, with realistic execution costs and additional low-risk factor controls, would address the principal remaining challenges.

V. CONCLUSION

The full model comparison in Table II establishes the broader allocation context. We conclude with the three rules examined throughout the subsequent comparisons: Market-cap weight, Equal weight, and constant-correlation GMVP (CC GMVP). They represent the original allocation, a benchmark requiring no return-moment estimates, and a structured minimum-variance allocation, respectively. S&P 500, CRSP, and FF remain market benchmarks rather than additional portfolio optimization methods. Common monthly constituents identify differences in weighting conditional on historical selection; they do not independently validate the selection rule.

At 20 basis points round trip, Market-cap weight has CAGR 14.05% and Sharpe 1.110, compared with 13.52% and 1.087 for Equal weight, and 13.20% and 1.069 for CC GMVP. The latter two have slightly lower full-sample volatility, but neither improves CAGR or Sharpe. Their mean-return differences from Market-cap weight are insignificant after the reported multiple-comparison adjustment. Thus, the reported return and Sharpe comparisons provide no statistically supported incremental advantage for either reweighting rule.

A. Selection Exposures and Regime Dependence

Market-cap weight and Equal weight require no estimated covariance or expected return. Their defensive performance is consistent with substantial risk reduction entering through the common low-beta and low-volatility selection. Equal weighting distributes capital uniformly across the selected names, while market-capitalization weighting retains size-related concentration subject to the position cap. Equal weight gains 6.66 cumulative-return points relative to Market-cap weight in Bull II, but trails it in Bull I, Shock I, and both crisis windows. This variation is consistent with changing returns to the exposures induced by weighting, rather than a stable advantage from avoiding covariance estimation.

TABLE IX
SELECTION COMPARISON UNDER THE ARTICLE'S RETURN AND COST CONVENTIONS. CAGR, ANNUALIZED VOLATILITY, AND MAXIMUM DRAWDOWN (MDD) ARE PERCENTAGES; SHARPE IS DIMENSIONLESS. SELECTION AND PORTFOLIO SIZE CHANGE TOGETHER. THESE ARE ALLOCATION BENCHMARKS, NOT A RE-ESTIMATION OF ALL OPTIMIZERS ON THE NEW UNIVERSE.

Selection (N)	Allocation	CAGR	Volatility	Sharpe	MDD
Defensive (30)	Equal weight	13.52	11.18	1.087	-30.77
Defensive (30)	Market-cap weight	14.05	11.39	1.110	-29.00
Return-oriented (50)	Equal weight	16.60	20.37	0.801	-49.18
Return-oriented (50)	Market-cap weight	16.48	18.45	0.858	-48.05

CC GMVP reinforces the marginal-risk component of selection through its common-correlation covariance model. Relative to Market-cap weight, its low-volatility score rises by 2.04 percentile points, its coskewness score falls by 1.06, and fitted beta falls from approximately 0.60 to 0.50. It gains 11.09 cumulative-return points in Recovery I and smaller amounts in Bull III and Pre-Crisis, but loses an additional 4.18 points in Crisis II. The model therefore does more than reduce estimated variance: it changes the balance among selection characteristics. Negative coskewness represents potential compensation for adverse-state exposure, not a direct hedge, and its premium is not separately identified here. The full-sample return shortfall arises mainly before costs, so additional trading alone does not explain CC GMVP's weaker growth.

The two crises illustrate why the evaluation criterion matters. In Crisis I, CC GMVP has lower volatility and a less severe MDD than Market-cap weight, yet a larger cumulative loss. All three selected portfolios outperform the market benchmarks in that episode, whereas all three lag them in Crisis II. The latter includes the V-shaped rebound, so it tests both loss containment and participation in recovery. A trailing covariance estimate and stronger low-risk tilt may respond poorly to such a transition, but the present results do not identify that mechanism causally. Neither a defensive selection label nor a minimum-variance objective guarantees superior terminal wealth in every crisis.

B. Market Evidence, Limitations, and Future Work

All three strategy CAGRs exceed the S&P 500 price return of 8.61% and the CRSP and FF total returns of 10.86% and 11.54%. The primary S&P comparison excludes benchmark dividends; CRSP and FF provide complementary comparisons with distributions included. Their constituent coverage also differs, so the three gaps are neither interchangeable nor independent evidence of a new premium.

For Market-cap weight, the FF-market Sharpe gap is 0.379 with 95% interval $[0.079, 0.720]$, and annualized FF5 alpha is 5.67% (HAC statistic 3.99). In 2012–2020 the gap falls to 0.038, with an interval containing zero and insignificant alpha. These nominal results concentrate the evidence in the earlier period. Stable strategy Sharpe does not rule out overfitting; historical specification search remains unaccounted for, and FF5 omits explicit BAB and low-volatility controls. Short, retrospectively defined regimes also limit inference about persistence or an implementable switching rule.

The separate return-oriented selection raises full-sample CAGR under Market-cap weight and Equal weight but increases volatility and drawdown and lowers Sharpe. Different stock counts and associated position limits prevent attributing those differences solely to the selection factors. More generally, missing upstream inputs restrict replication to allocation and accounting conditional on supplied constituents, and constant proportional costs do not establish executable capacity.

Future work should first reconstruct point-in-time selection inputs and compare these three weighting rules across characteristic ablations and both selection strategies under common stock counts, position limits, costs, and evaluation dates. A common ex-ante risk budget would help separate higher growth obtained through greater risk from improved allocation efficiency. Tracking beta, volatility, and coskewness exposures alongside factor-return contributions would test whether CC GMVP's stronger low-risk tilt explains its regime outcomes. BAB and low-volatility controls would assess whether alpha reflects known low-risk effects; covariance forecast evaluation and turnover-matched comparisons would distinguish estimation and trading channels.

Estimation windows and any exposure constraints should be chosen within training and validation samples before evaluation on previously unused data. Any adaptive choice among Market-cap weight, Equal weight, and CC GMVP must use states observable at the decision date, rather than the current ex-post regime calendar. Finally, spreads, liquidity-dependent impact, and capacity should enter a common execution model. These extensions would test when CC GMVP adds value beyond the selected stocks and the two simple weighting benchmarks.

APPENDIX A
MODEL SPECIFICATIONS AND NUMERICAL VALIDATION

A. Model Specifications

Table X summarizes the statistical inputs and solution procedures for the 19 allocation rules. The 36-month estimation window, 30-stock portfolio, $2/N$ position cap, ridge coefficient 0.3, 50:50 mixture, and risk-aversion coefficient 5 are study-specific parameter choices; the cited theories do not establish their empirical optimality.

B. Numerical Accuracy

Independent calculations assess feasibility and optimality across 4,085 allocations (19 rules over 215 months). The largest budget residual is 2.66×10^{-15} and the largest position-bound violation is 6.15×10^{-12}. Across 1,720 covariance estimates, the maximum relative discrepancy from independently evaluated formulas is 1.37×10^{-15}. Sample and norm-constrained

TABLE X
COMPLETE MAPPING OF THE 19 ALLOCATION RULES TO THEIR MATHEMATICAL AND NUMERICAL SPECIFICATIONS. THE SAME CONSTITUENT SET AND FEASIBLE REGION APPLY THROUGHOUT.

Rule	Statistical input or construction	Computation
Market-cap weight	Recorded historical weights	Rounding projection
Equal weight	$w_i = 1/N$	Direct assignment
Sample GMVP	Sample covariance S	Algorithms 5 and 2
Constant-correlation GMVP	Common-correlation F	Algorithms 5 and 2
LW constant-correlation GMVP	Estimated shrinkage toward F_0	Algorithms 5 and 2
LW identity GMVP	Estimated shrinkage toward mI	Algorithms 5 and 2
RMT GMVP	Filtered correlation spectrum	Algorithms 5 and 2
QIS GMVP	Nonlinear sample-eigenvalue correction	Algorithms 5 and 2
Ridge GMVP	$S + 0.3\,\mathrm{tr}(S)I/N$	Algorithms 5 and 2
FF5 GMVP	Factor covariance plus diagonal residuals	Algorithms 5 and 2
Inverse volatility	Normalized $S_{ii}^{-1/2}$	Capped-simplex projection
Inverse variance	Normalized S_{ii}^{-1}	Capped-simplex projection
Constrained risk budgeting	Equal risk budgets with explicit box	Algorithm 4
Maximum diversification	Marginal volatility numerator, S	Algorithm 3
Bayes–Stein Sharpe	Shrunk mean, LW identity covariance	Algorithm 3
Minimum CVaR	Historical losses, tail probability 0.10	Algorithm 6
GMVP–$1/N$ mixture	Fixed convex combination, coefficient 0.5	Algorithm 2 then mixing
Norm GMVP	$\|w\|_2^2 \le 1/15$, redundant here	Sample GMVP solution
Transaction-cost GMVP	S, pretrade weights, $\lambda = 0.0004$	Algorithm 2 with joint proximal map

GMVP weights coincide, as implied by the redundant norm constraint.

The maximum normalized objective-error bounds are 3.49×10^{-9} for GMVP and 8.19×10^{-10} for transaction-cost GMVP; the CVaR primal–dual gap is at most 1.39×10^{-16}. Optimality conditions are checked for maximum diversification, Bayes–Stein Sharpe, and constrained risk budgeting in every estimation window. For the six negative-numerator Sharpe problems, exhaustive evaluation of all $\binom{30}{15} = 155{,}117{,}520$ vertices per window confirms the solutions. Appendix B gives the bounds used in ratio optimization. These results establish numerical accuracy conditional on the estimated inputs; they do not address estimation uncertainty or economic performance.

Independent reconstruction of monthly net returns agrees within 4.17×10^{-16} at baseline and 5.28×10^{-16} across all six transaction-cost scenarios. Performance measures, regime aggregation, HAC tests, Holm adjustments, and the paired bootstrap are independently recalculated. Each market benchmark is checked against its source series using the same evaluation dates.

APPENDIX B
CERTIFICATES FOR RATIO OPTIMIZATION

For the positive-numerator quadratic program, write $Cy \ge 0$ for the lower and transformed upper bounds. Its Lagrangian is $y'\Sigma y + \nu(a'y - 1) - \eta' Cy$, with $\eta \ge 0$. Define $h = \nu a - C'\eta$. Minimization over y gives the dual lower bound

$$d(\nu, \eta) = -\tfrac{1}{4} h' \Sigma^{-1} h - \nu. \tag{42}$$

Thus any dual-feasible multipliers bound the optimum, even when stationarity is approximate. The implementation accepts the solution only when primal feasibility is within 10^{-8} and the relative primal–dual objective gap is at most 10^{-8}. Objective and covariance scaling are used for conditioning and do not change the portfolio.

For the strictly negative-numerator branch, put $b = -a$ and work with binary vertex coordinates $z = w/u$ for the present reciprocal cap. At a search node, A is the set of already selected coordinates, U is the undecided set, and k more coordinates must be selected. Write $b_A = \sum_{i \in A} b_i$ and $q_A = \sum_{i,j \in A} \Sigma_{ij}$. For each $j \in U$, let

$$d_j = \Sigma_{jj} + 2 \sum_{i \in A} \Sigma_{ij} + \sum_{\substack{k-1 \text{ largest entries} \\ \ell \in U \setminus \{j\}}} \Sigma_{j\ell}. \tag{43}$$

For every completion $B \subset U$ of size k, $z'\Sigma z \le q_A + \sum_{j \in B} d_j$. This remains an upper bound when some covariances are negative. For any real v, convexity of the square gives $(b_A + \sum_{j \in B} b_j)^2 \ge 2v(b_A + \sum_{j \in B} b_j) - v^2$. Consequently a lower bound on $(b'z)^2 - \tau^2 z'\Sigma z$ is

$$L(v) = 2v b_A - v^2 - \tau^2 q_A + \sum_{k \text{ smallest } j \in U} (2v b_j - \tau^2 d_j). \tag{44}$$

Here $\tau = (1 - 10^{-10})$ times the incumbent value of $b'z/\sqrt{z'\Sigma z}$. A nonnegative lower bound, enlarged by a floating-point error margin in the pruning test, excludes an improvement exceeding this tolerance. Feasible one-coordinate swaps supply an incumbent but never justify pruning. The algorithm terminates without accepting a solution if its node limit is reached.

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